Notes on Varieties of Semiring Solvers

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Contents

1 Preliminaries

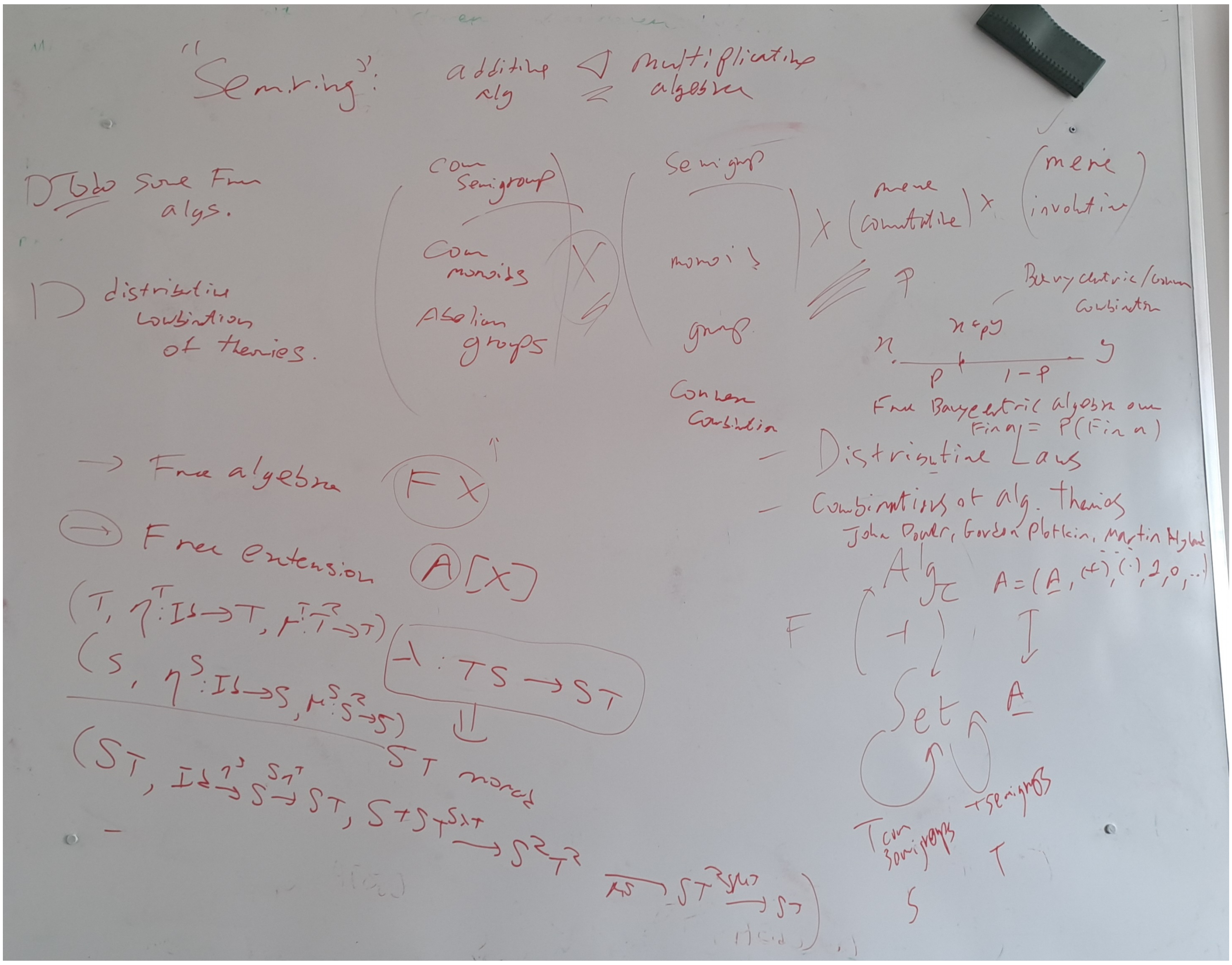

Figure 1: Blackboard at the start of the internship.

For convenience, we recall some definitions we'll be using.

1.1 Algebraic Structures

Definition 1 (Semigroup)

A *semigroup* is a set S equipped with an associative binary operation $\cdot : S \times S \to S$ such that for all $a, b, c \in S$:

- $a \cdot (b \cdot c) = (a \cdot b) \cdot c$ (associativity)

Definition 2 (Monoid)

A *monoid* is a set M equipped with an associative binary operation $\cdot : M \times M \to M$ and an identity element $e \in M$ such that for all $a, b, c \in M$:

- $a \cdot (b \cdot c) = (a \cdot b) \cdot c$ (associativity)
- $e \cdot a = a \cdot e = a$ (identity element)

Definition 3 (Group)

A *group* is a set G equipped with an associative binary operation $\cdot : G \times G \to G$, an identity element $e \in G$, and an inverse operation such that for all $a, b, c \in G$:

- $a \cdot (b \cdot c) = (a \cdot b) \cdot c$ (associativity)
- $e \cdot a = a \cdot e = a$ (identity element)
- For each $a \in G$, there exists an element $a^{-1} \in G$ such that $a \cdot a^{-1} = a^{-1} \cdot a = e$ (inverse element)

A group is called *abelian* if its operation is commutative, i.e., for all $a, b \in G$, $a \cdot b = b \cdot a$.

Definition 4 (Semiring)

A *semiring* is a set R equipped with two binary operations: addition $+$ and multiplication $\cdot$, such that:

- $(R, +)$ is a commutative monoid with identity element 0.
- $(R, \cdot)$ is a monoid with identity element 1.
- Multiplication distributes over addition from both sides: for all $a, b, c \in R$,

$$a \cdot (b + c) = a \cdot b + a \cdot c,$$
$$(a + b) \cdot c = a \cdot c + b \cdot c.$$

- The additive identity 0 is an absorbing element for multiplication: for all $a \in R$, $0 \cdot a = a \cdot 0 = 0$.

1.2 Categories

Definition 5 (Category)

A *category* $\mathcal{C}$ consists of:

- A class of objects $\mathrm{Ob}(\mathcal{C})$,
- For each pair of objects $a, b \in \mathrm{Ob}(\mathcal{C})$, a set of morphisms $\mathrm{Hom}_{\mathcal{C}}(a, b)$,
- For each object $a \in \mathrm{Ob}(\mathcal{C})$, an identity morphism $\mathrm{id}_a \in \mathrm{Hom}_{\mathcal{C}}(a, a)$,
- A composition operation $\circ$ such that for morphisms $f : a \to b$ and $g : b \to c$, there is a morphism $g \circ f : a \to c$,

satisfying the following axioms:

- Associativity: For morphisms $f : a \to b$, $g : b \to c$, and $h : c \to d$, we have $h \circ (g \circ f) = (h \circ g) \circ f$.
- Identity: For any morphism $f : a \to b$, we have $\mathrm{id}_b \circ f = f = f \circ \mathrm{id}_a$.

Example 6 (Set Category)

The category **Set** has sets as objects and functions as morphisms. The identity morphism for a set a is the identity function $\mathrm{id}_a : a \to a$ defined by $\mathrm{id}_a(x) = x$ for all $x \in a$. Composition of morphisms is given by function composition.

Definition 7 (Product)

Given two objects a and b in a category $\mathcal{C}$, a *product* of a and b is an object $a \times b$ together with two morphisms $\pi_a : a \times b \to a$ and $\pi_b : a \times b \to b$ such that for any object c and morphisms $f : c \to a$ and $g : c \to b$, there exists a unique morphism $\langle f, g \rangle : c \to a \times b$ making the following diagram commute:

$$\begin{tikzcd} & c \arrow[dl, "f"'] \arrow[d, dashed, "\langle f,g \rangle"] \arrow[dr, "g"] & \\ a & a \times b \arrow[l, "\pi_a"] \arrow[r, "\pi_b"'] & b \end{tikzcd}$$

Definition 8 (Coproduct)

Given two objects a and b in a category $\mathcal{C}$, a *coproduct* of a and b is an object $a + b$ together with two morphisms $\iota_a : a \to a + b$ and $\iota_b : b \to a + b$ such that for any object c and morphisms $f : a \to c$ and $g : b \to c$, there exists a unique morphism $[f, g] : a + b \to c$ making the following diagram commute:

$$\begin{tikzcd} a \arrow[r, "\iota_a"] \arrow[dr, "f"'] & a + b \arrow[d, dashed, "{[f,g]}"] & b \arrow[l, "\iota_b"'] \arrow[dl, "g"] \\ & c & \end{tikzcd}$$

Definition 9 (Monoidal Category)

A *monoidal category* is a category $\mathcal{C}$ equipped with a bifunctor $\otimes : \mathcal{C} \times \mathcal{C} \to \mathcal{C}$, an object I called the unit object, and natural isomorphisms (called associators and unitors) that satisfy certain coherence conditions. Specifically, for all objects a, b, c in $\mathcal{C}$, there are natural isomorphisms:

- Associator: $\alpha_{a,b,c} : (a \otimes b) \otimes c \to a \otimes (b \otimes c)$,
- Left unitor: $\lambda_a : I \otimes a \to a$,
- Right unitor: $\rho_a : a \otimes I \to a$,

such that the following diagrams commute:

$$\begin{tikzcd} ((a \otimes b) \otimes c) \otimes d \arrow[r, "\alpha_{a,b,c} \otimes 1_d"] & (a \otimes (b \otimes c)) \otimes d \arrow[r, "\alpha_{a,b \otimes c,d}"] & a \otimes ((b \otimes c) \otimes d) \\ (a \otimes b) \otimes (c \otimes d) \arrow[u, "\alpha_{a,b,c \otimes d}"] & & a \otimes (b \otimes (c \otimes d)) \arrow[u, "1_a \otimes \alpha_{b,c,d}"'] \end{tikzcd}$$

$$(I \otimes a) \otimes b \xrightarrow{\alpha_{I,a,b}} I \otimes (a \otimes b) \xleftarrow{\rho_a} a = a \otimes I \xrightarrow{\lambda_a} a \otimes I \xrightarrow{\alpha_{a,I,I}} (a \otimes I) \otimes I$$

Definition 10 (Functor)

Given two categories $\mathcal{C}$ and $\mathcal{D}$, a *functor* $F : \mathcal{C} \to \mathcal{D}$ consists of:

- A mapping $F : \mathrm{Ob}(\mathcal{C}) \to \mathrm{Ob}(\mathcal{D})$ that assigns to each object a in $\mathcal{C}$ an object $F(a)$ in $\mathcal{D}$,
- For each pair of objects $a, b \in \mathrm{Ob}(\mathcal{C})$, a mapping $F : \mathrm{Hom}_{\mathcal{C}}(a,b) \to \mathrm{Hom}_{\mathcal{D}}(F(a), F(b))$ that assigns to each morphism $f : a \to b$ in $\mathcal{C}$ a morphism $F(f) : F(a) \to F(b)$ in $\mathcal{D}$,

such that for all objects a, b, c in $\mathcal{C}$ and morphisms $f : a \to b$, $g : b \to c$, the following conditions hold:

- $F(g \circ f) = F(g) \circ F(f)$ (preservation of composition),
- $F(\mathrm{id}_a) = \mathrm{id}_{F(a)}$ (preservation of identity).

For a category $\mathcal{C}$, the *identity functor* $\mathrm{Id}_{\mathcal{C}} : \mathcal{C} \to \mathcal{C}$ is defined by $\mathrm{Id}_{\mathcal{C}}(a) = a$ for all objects a and $\mathrm{Id}_{\mathcal{C}}(f) = f$ for all morphisms f.

- A functor $F : \mathcal{C} \to \mathcal{D}$ is *faithful* if, for every pair of objects $a, b \in \mathrm{Ob}(\mathcal{C})$, the map $F : \mathrm{Hom}_{\mathcal{C}}(a,b) \to \mathrm{Hom}_{\mathcal{D}}(Fa, Fb)$ is injective.
- A functor $F : \mathcal{C} \to \mathcal{D}$ is *full* if, for every pair of objects $a, b \in \mathrm{Ob}(\mathcal{C})$, the map $F : \mathrm{Hom}_{\mathcal{C}}(a,b) \to \mathrm{Hom}_{\mathcal{D}}(Fa, Fb)$ is surjective.
- A functor that is both full and faithful is called *fully faithful.*

Notation: We often denote the image of an object a under a functor F as Fa instead of $F(a)$ and the image of a morphism $f : a \to b$ as $Ff : Fa \to Fb$.

Example 11 (Diagonal functor)

Given a category $\mathcal{C}$, the *diagonal functor* $\Delta : \mathcal{C} \to \mathcal{C} \times \mathcal{C}$ is defined by $\Delta(a) = (a, a)$ for each object a in $\mathcal{C}$ and $\Delta(f) = (f, f)$ for each morphism $f : a \to b$ in $\mathcal{C}$.

Definition 12 (Product preserving functor)

A functor $F : \mathcal{C} \to \mathcal{D}$ is said to *preserve products* if for every collection of objects $\{a_i\}_{i \in I}$ in $\mathcal{C}$, the natural morphism from $F(\prod_{i \in I} a_i)$ to $\prod_{i \in I} F(a_i)$ is an isomorphism in $\mathcal{D}$. In other words, F preserves the structure of products up to isomorphism.

Definition 13 (Natural Transformation)

Given two functors $F, G : \mathcal{C} \to \mathcal{D}$ between categories $\mathcal{C}$ and $\mathcal{D}$, a *natural transformation* $\eta : F \to G$ consists of a family of morphisms $\{\eta_a : Fa \to Ga\}_{a \in \mathrm{Ob}(\mathcal{C})}$ such that for every morphism $f : a \to b$ in $\mathcal{C}$, the following diagram commutes:

$$\begin{tikzcd} Fa \arrow[r, "Ff"] \arrow[d, "\eta_a"'] & Fb \arrow[d, "\eta_b"] \\ Ga \arrow[r, "Gf"'] & Gb \end{tikzcd}$$

Definition 14 (Adjunction)

Given two categories $\mathcal{C}$ and $\mathcal{D}$, a functor $F : \mathcal{C} \to \mathcal{D}$ is a *left adjoint* to a functor $G : \mathcal{D} \to \mathcal{C}$ (and G is a *right adjoint* to F) if there is a natural isomorphism:

$$\mathrm{Hom}_{\mathcal{D}}(Fa, b) \cong \mathrm{Hom}_{\mathcal{C}}(a, Gb)$$

for all objects a in $\mathcal{C}$ and b in $\mathcal{D}$. This means that for every morphism $f : Fa \to b$ in $\mathcal{D}$, there is a unique morphism $g : a \to Gb$ in $\mathcal{C}$ such that the following diagram commutes:

$$\begin{tikzcd}
Fa \arrow[r, "f"] & b \arrow[d, "G"] \\
a \arrow[u, "F"] \arrow[r, "g"'] & Gb
\end{tikzcd}$$

We often denote this adjunction by $F \dashv G$.

There are actually several different but equivalent ways to define adjunctions. They are linked thanks to the Yoneda lemma.

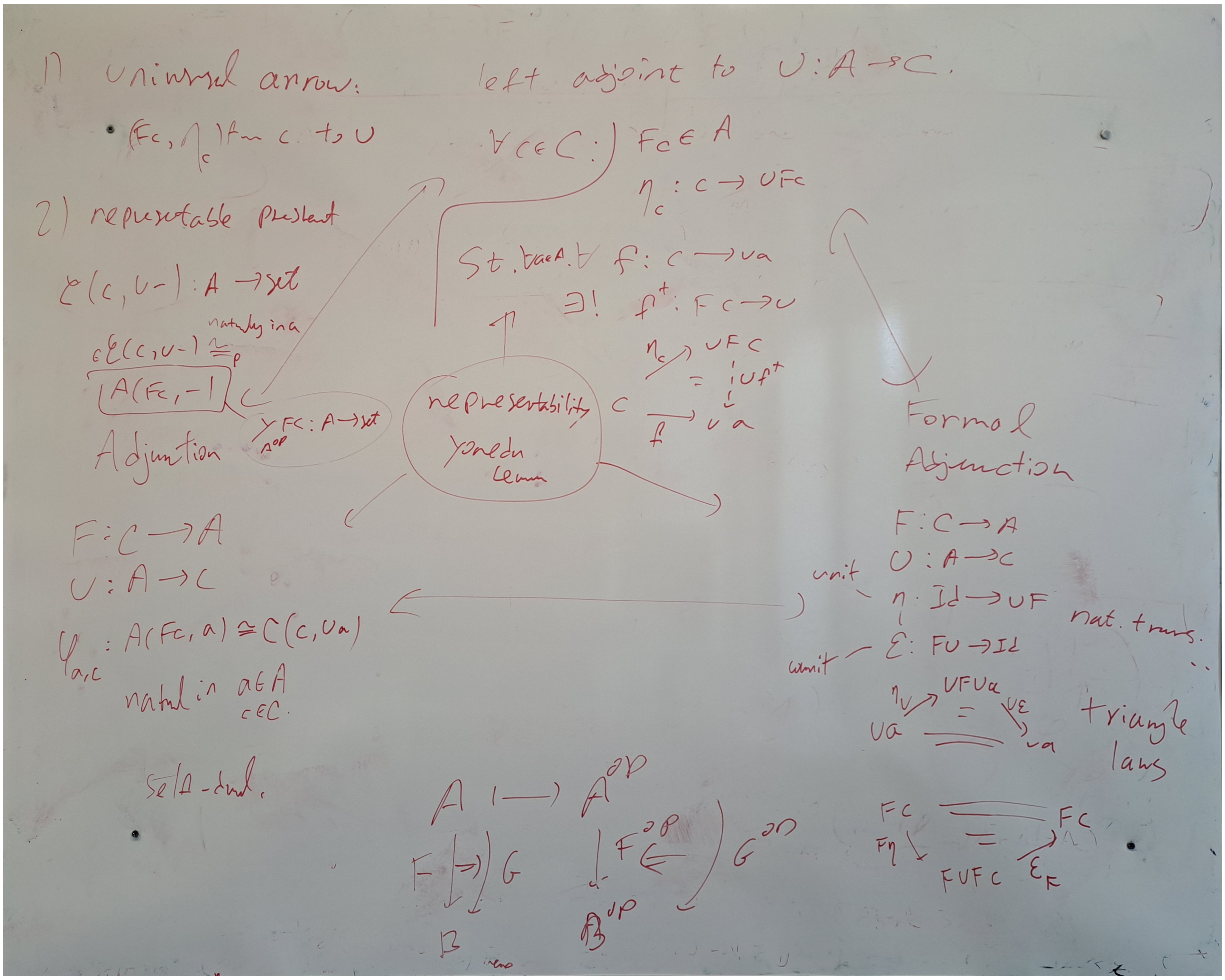

Figure 2: Different equivalent definitions of adjunctions.

One important definition is that of the universal arrow from MacLane's *Categories for the Working Mathematician* [Mac71].

Definition 15 (Universal arrow)

Let $U : A \to C$ be a functor, and let $c \in C$.

An *universal arrow* from c to U is a pair (Fc, η_c) where:

- $Fc \in A$
- $\eta_c : c \to UFc$ is a morphism in C

such that for every $a \in A$, and every $f : c \to Ua$ in C, there exists a unique morphism $f^\dagger : Fc \to a$ in A such that the following diagram commutes:

$$\begin{tikzcd} c \arrow[r, "\eta_c"] \arrow[dr, "f"'] & UFc \arrow[d, dashed, "Uf^\dagger"] \\ & Ua \end{tikzcd} \qquad \begin{tikzcd} Fc \arrow[d, "f^\dagger"] \\ a \end{tikzcd}$$

We can dualize this definition to get the notion of a universal arrow from U to c.

A universal arrow from U to c is a pair (Fc, ε_c) where:

- $Fc \in A$
- $\varepsilon_c : UFc \to c$ is a morphism in C

such that for every $a \in A$, and every $f : Ua \to c$ in C, there exists a unique morphism $f^\dagger : a \to Fc$ in A such that the following diagram commutes:

$$\begin{tikzcd} c & UFc \arrow[l, "\varepsilon_c"'] \\ & Ua \arrow[ul, "f"] \arrow[u, dashed, "Uf^\dagger"'] \end{tikzcd} \qquad \begin{tikzcd} Fc \\ a \arrow[u, "f^\dagger"'] \end{tikzcd}$$

Example 16 (Adjunction from Universal Arrows)

Given a functor $U : A \to C$ such that for every object $c \in C$, there exists a universal arrow from c to U, we can construct an adjunction $F \dashv U$ where:

- The functor $F : C \to A$ is defined by sending each object c in C to the object Fc in A that appears in the universal arrow from c to U, and sending each morphism $f : c \to c'$ in C to the unique morphism $Ff : Fc \to Fc'$ in A that makes the corresponding diagram commute.
- The unit $\eta : \mathrm{Id}_C \to UF$ is given by the family of morphisms $\{\eta_c : c \to UFc\}_{c \in C}$ that are part of the universal arrows.

Universal arrows have more uses than just defining adjunctions. Here are other examples.

Example 17 (Defining the coproduct with an universal arrow)

Let A be a category, and let $a, b \in A$. A *coproduct* of a and b is a universal arrow from (a, b) to the diagonal functor $\Delta : A \to A \times A$, where $\Delta(c) = (c, c)$ for any object $c \in A$.

$$\begin{tikzcd} a \arrow[r, "\iota_a"] \arrow[dr, "f"'] & a+b \arrow[d, dashed, "{[f,g]}"] & b \arrow[l, "\iota_b"'] \arrow[dl, "g"] \\ & c & \end{tikzcd} \quad \sim \quad \begin{tikzcd} (a,b) \arrow[r, "\eta_{(a,b)}"] \arrow[dr, "{(f,g)}"'] & \Delta(a+b) \arrow[d, dashed, "{\Delta[f,g]}"] \\ & \Delta c \end{tikzcd}$$

We have $\eta_{(a,b)} = (\iota_a, \iota_b)$.

Example 18 (Defining the product with an universal arrow)

Let A be a category, and let $a, b \in A$. A *product* of a and b is a universal arrow from the diagonal functor $\Delta : A \to A \times A$ to (a, b).

$$\begin{tikzcd} & c \arrow[dl, "f"'] \arrow[d, dashed, "\langle f,g \rangle"] \arrow[dr, "g"] & \\ a & a \times b \arrow[l, "\pi_a"] \arrow[r, "\pi_b"'] & b \end{tikzcd} \quad \sim \quad \begin{tikzcd} \Delta c \arrow[r, "\varepsilon_{(a,b)}"] \arrow[dr, "(f,g)"'] & \Delta(a \times b) \arrow[d, dashed, "(\pi_a,\pi_b)"] \\ & (a,b) \end{tikzcd}$$

We have $\varepsilon_{(a,b)} = \langle f, g \rangle$.

Definition 19 (Pullback of two morphisms)

Let $f : X \to Z$ and $g : Y \to Z$ be two morphisms in a category $\mathcal{C}$. A *pullback* of f and g consists of an object P and two morphisms $p_1 : P \to X$ and $p_2 : P \to Y$ such that the following diagram commutes:

$$\begin{tikzcd} P \arrow[r, "p_2"] \arrow[d, "p_1"'] & Y \arrow[d, "g"] \\ X \arrow[r, "f"'] & Z \end{tikzcd}$$

Moreover, for any triple (Q, q_1, q_2) where $q_1 : Q \to X$ and $q_2 : Q \to Y$ are morphisms with $fq_1 = gq_2$, there must exist $u : Q \to P$ such that $p_1 \circ u = q_1$ and $p_2 \circ u = q_2$.
The situation is illustrated in the following commutative diagram:

$$\begin{tikzcd} Q \arrow[dr, dashed, "u"] \arrow[drr, bend left, "q_2"] \arrow[ddr, bend right, "q_1"'] & & \\ & P \arrow[r, "p_2"] \arrow[d, "p_1"'] & Y \arrow[d, "g"] \\ & X \arrow[r, "f"'] & Z \end{tikzcd}$$

This condition can be reformulated with universal arrows as follows: a pullback of f and g is a universal arrow from the functor $\Delta : \mathcal{C} \to \mathcal{C} \times \mathcal{C}$ to the object (f, g) in $\mathcal{C} \times \mathcal{C}$.

Example 20 (Pullback in the category of categories)

Let $\mathcal{A}, \mathcal{B}, \mathcal{Z} \in \mathbf{Set}$ such that we have functors $F : \mathcal{A} \to \mathcal{Z}$ and $G : \mathcal{B} \to \mathcal{Z}$.

The pullback of F and G is the category $\mathcal{P}$ defined as follows:

- The objects of $\mathcal{P}$ are pairs (a, b) where a is an object of $\mathcal{A}$, b is an object of $\mathcal{B}$, and $Fa = Gb$.
- The morphisms of $\mathcal{P}$ from (a, b) to (a', b') are pairs (f, g) where $f : a \to a'$ is a morphism in $\mathcal{A}$, $g : b \to b'$ is a morphism in $\mathcal{B}$, and $Ff = Gg$.
- The composition of morphisms in $\mathcal{P}$ is defined componentwise: if $(f, g) : (a, b) \to (a', b')$ and $(f', g') : (a', b') \to (a'', b'')$ are morphisms in $\mathcal{P}$, then their composition is given by $(f' \circ f, g' \circ g) : (a, b) \to (a'', b'')$.
- The identity morphism on an object (a, b) in $\mathcal{P}$ is given by $(\mathrm{id}_a, \mathrm{id}_b)$, where id_a is the identity morphism on a in $\mathcal{A}$ and id_b is the identity morphism on b in $\mathcal{B}$.

Proof
Consider the following commuting diagram:

$$\begin{tikzcd}
\mathcal{Q} \arrow[dr, dashed, "U"] \arrow[drr, bend left, "Q_2"] \arrow[ddr, bend right, "Q_1"'] & & \\
& \mathcal{P} \arrow[r, "\pi_2"] \arrow[d, "\pi_1"'] & \mathcal{B} \arrow[d, "G"] \\
& \mathcal{A} \arrow[r, "F"'] & \mathcal{Z}
\end{tikzcd}$$

We need to find U and to prove that it is unique up to isomorphism.

We define U as follows: for any object q in $\mathcal{Q}$, we have $Q_1q \in \mathcal{A}$ and $Q_2q \in \mathcal{B}$. Since $FQ_1q = GQ_2q$, we can define $Uq = (Q_1q, Q_2q)$. For any morphism $m : q \to q'$ in $\mathcal{Q}$, we have $Q_1m : Q_1q \to Q_1q'$ and $Q_2m : Q_2q \to Q_2q'$. Since $FQ_1m = GQ_2m$, we can define $Um = (Q_1m, Q_2m)$.

It is straightforward to check that U is a functor and that the diagram commutes.

Moreover, if $U' : \mathcal{Q} \to \mathcal{P}$ is another functor such that the diagram commutes, then for any object q in $\mathcal{Q}$, we have $U'q = (Q_1q, Q_2q) = Uq$. For any morphism $m : q \to q'$ in $\mathcal{Q}$, we have $U'm = (Q_1m, Q_2m) = Um$. Hence, U is unique up to isomorphism. □

1.3 Monads

Definition 21 (Monad)

A *monad* on a category $\mathcal{C}$ is a triple (T, η, μ) where:

- $T : \mathcal{C} \to \mathcal{C}$ is a functor,
- $\eta : \mathrm{Id}_{\mathcal{C}} \to T$ is a natural transformation called the unit,
- $\mu : T^2 \to T$ is a natural transformation called the multiplication,

such that the following diagrams commute:

$$\begin{tikzcd}
T \arrow[r, "T\eta"] \arrow[d, "\eta T"'] \arrow[dr, equal] & T^2 \arrow[d, "\mu"] \\
T^2 \arrow[r, "\mu"'] & T
\end{tikzcd}
\qquad
\begin{tikzcd}
T^3 \arrow[r, "T\mu"] \arrow[d, "\mu T"'] & T^2 \arrow[d, "\mu"] \\
T^2 \arrow[r, "\mu"'] & T
\end{tikzcd}$$

Example 22 (Monad from an Adjunction)

Given an adjunction $F \dashv G$ between categories $\mathcal{C}$ and $\mathcal{D}$, we can construct a monad on $\mathcal{C}$ as follows:

- The functor $T : \mathcal{C} \to \mathcal{C}$ is defined by $T = GF$.
- The unit $\eta : \mathrm{Id}_{\mathcal{C}} \to T$ is given by the natural transformation that assigns to each object a in $\mathcal{C}$ the morphism $\eta_a : a \to GFa$ corresponding to the identity morphism on Fa under the adjunction.
- The multiplication $\mu : T^2 \to T$ is given by the natural transformation that assigns to each object a in $\mathcal{C}$ the morphism $\mu_a : GFGFa \to GFa$ corresponding to the composition of

morphisms in $\mathcal{D}$ under the adjunction.

Definition 23 (Distributive law between monads)

Given two monads (T, η^T, μ^T) and (S, η^S, μ^S) on a category $\mathcal{C}$, a *distributive law* of T over S is a natural transformation $\lambda : TS \to ST$ such that the following diagrams commute:

$$\begin{tikzcd} & T \arrow[dl, "T\eta^S"'] \arrow[dr, "\eta^S T"] & \\ TS \arrow[rr, "\lambda"'] & & ST \end{tikzcd} \qquad \begin{tikzcd} & S \arrow[dl, "\eta^T S"'] \arrow[dr, "S\eta^T"] & \\ TS \arrow[rr, "\lambda"'] & & ST \end{tikzcd}$$

$$\begin{tikzcd} T^2S \arrow[r, "T\lambda"] \arrow[d, "\mu^T S"'] & TST \arrow[r, "\lambda T"] & ST^2 \arrow[d, "S\mu^T"] \\ TS \arrow[rr, "\lambda"'] & & ST \end{tikzcd} \qquad \begin{tikzcd} TS^2 \arrow[r, "\lambda S"] \arrow[d, "T\mu^S"'] & STS \arrow[r, "S\lambda"] & S^2T \arrow[d, "\mu^S T"] \\ TS \arrow[rr, "\lambda"'] & & ST \end{tikzcd}$$

This law induces a composite monad ST on $\mathcal{C}$ with

- unit $\eta^{ST} = \eta^S \circ \eta^T$
- multiplication $STST \xrightarrow{S\lambda T} SSTT \xrightarrow{\mu^S \mu^T} ST$

Definition 24 (Algebra over a monad)

Let (T, η, μ) be a monad on a category $\mathcal{C}$.

An *algebra over the monad* T (or simply a T-algebra) consists of:

- An object A in $\mathcal{C}$,
- A morphism $a : TA \to A$ in $\mathcal{C}$, called the structure map, such that the following diagrams commute:

$$\begin{tikzcd} A \arrow[r, "\eta_A"] \arrow[dr, "\mathrm{id}_A"'] & TA \arrow[d, "a"] \\ & A \end{tikzcd} \qquad \begin{tikzcd} T^2A \arrow[r, "Ta"] \arrow[d, "\mu_A"'] & TA \arrow[d, "a"] \\ TA \arrow[r, "a"'] & A \end{tikzcd}$$

The left one is called the unit property, and the right one is called the action property.

We'll now reformulate the definition of monads in terms of Kleisli categories and extension operators.

Definition 25 (Kleisli category)

Given a monad (T, η, μ) on a category $\mathcal{C}$, the *Kleisli category* $\mathcal{C}_T$ has:

- The same objects as $\mathcal{C}$,
- For each pair of objects a, b, the morphisms from a to b in $\mathcal{C}_T$ are given by $\mathrm{Hom}_{\mathcal{C}_T}(a, b) = \mathrm{Hom}_{\mathcal{C}}(a, Tb)$,
- The identity morphism on an object a is given by $\eta_a : a \to Ta$,
- The composition of morphisms $f : a \to Tb$ and $g : b \to Tc$ in $\mathcal{C}_T$ is given by the morphism $g \circ_T f : a \to Tc$ defined as the composition:

$$a \xrightarrow{f} Tb \xrightarrow{Tg} T^2c \xrightarrow{\mu_c} Tc$$

Definition 26 (Extension operator)

Given a monad (T, η, μ) on a category $\mathcal{C}$, the *extension operator* $(-)^\sharp$ is defined for each morphism $f : a \to Tb$ in $\mathcal{C}$ as the morphism $f^\sharp : Ta \to Tb$ given by the composition:

$$Ta \xrightarrow{Tf} T^2 b \xrightarrow{\mu_b} Tb$$

In other words, $f^\sharp = \mu_b \circ Tf$.

We have the following properties of the extension operator:

- $\eta_a^\sharp = \mathrm{id}_{Ta}$ for all objects a in $\mathcal{C}$.
- For any morphism $f : a \to Tb$, we have $f^\sharp \circ \eta_a = f$.
- For any morphisms $f : a \to Tb$ and $g : b \to Tc$, we have $(g^\sharp \circ f)^\sharp = g^\sharp \circ f^\sharp$.

Definition 27 (Kleisli triple)

A *Kleisli triple* on a category $\mathcal{C}$ consists of:

- A mapping $T : \mathrm{Ob}(\mathcal{C}) \to \mathrm{Ob}(\mathcal{C})$,
- For each object a in $\mathcal{C}$, a morphism $\eta_a : a \to Ta$,
- For each morphism $f : a \to Tb$ in $\mathcal{C}$, a morphism $f^\sharp : Ta \to Tb$,

such that the following conditions hold:

- $\eta_a^\sharp = \mathrm{id}_{Ta}$ for all objects a in $\mathcal{C}$.
- the following diagrams commute

$$\begin{tikzcd} a \arrow[r, "\eta_a"] \arrow[dr, "f"'] & Ta \arrow[d, "f^\sharp"] \\ & Tb \end{tikzcd} \qquad \begin{tikzcd} Ta \arrow[r, "f^\sharp"] \arrow[dr, "(g^\sharp \cdot f)^\sharp"'] & Tb \arrow[d, "g^\sharp"] \\ & Tc \end{tikzcd}$$

Proposition 28

Given a category $\mathcal{C}$, there is a one-to-one correspondence between monads on $\mathcal{C}$ and Kleisli triples on $\mathcal{C}$.

Proof

Given a monad (T, η, μ) on $\mathcal{C}$, we can construct a Kleisli triple by defining the extension operator as $f^\sharp = \mu_b \circ Tf$ for each morphism $f : a \to Tb$.

Conversely, given a Kleisli triple $(T, \eta, (-)^\sharp)$ on $\mathcal{C}$, we can construct a monad by defining the multiplication $\mu_a : T^2 a \to Ta$ as $\mu_a = (\mathrm{id}_{Ta})^\sharp$ for each object a in $\mathcal{C}$.
The conditions of the Kleisli triple ensure that the monad axioms are satisfied, and the construction is reversible, establishing the one-to-one correspondence. □

Remark : Kleisli triples are sometimes referred to as *monads in extensive form.*

Example 29 (Monads in terms of binds and returns)

In the context of programming languages, a monad can be described in terms of a type constructor T and two operations: `return` (or `unit`) and `bind`. The `return` operation corresponds to the unit

η of the monad, while the `bind` operation corresponds to the extension operator $(-)^\sharp$. Specifically, for a monad (T, η, μ) on a category $\mathcal{C}$, we can define:

- $\texttt{return}_a = \eta_a : a \to Ta$ for each object a in $\mathcal{C}$,
- $\texttt{bind}_{a,b}(f) = f^\sharp : Ta \to Tb$ for each morphism $f : a \to Tb$ in $\mathcal{C}$.

We use the notion of $\mathbb{S}$-algebra of [MW10] to define the distributive laws for monads in extensive form.

Definition 30 ($\mathbb{S}$-algebra, [MW10])

Given a monad in extensive form $\mathbb{S} = (S, \eta, (-)^{\mathbb{S}})$ on a category $\mathcal{C}$, an $\mathbb{S}$-*algebra* is a pair $\mathbb{B} = (B, (-)^{\mathbb{B}})$ where:

- $B \in \mathcal{C}$
- $(-)^{\mathbb{B}}_X : \mathcal{C}(X, B) \to \mathcal{C}(SX, B)$

such that the following diagrams commute for every $h : X \to B$ and $k : Y \to SX$:

$$\begin{tikzcd} X \arrow[r, "\eta X"] \arrow[dr, "h"'] & SX \arrow[d, "h^{\mathbb{B}}"] \\ & B \end{tikzcd} \qquad \begin{tikzcd} SY \arrow[r, "k^{\mathbb{S}}"] \arrow[dr, "(h^{\mathbb{B}}k)^{\mathbb{B}}"'] & SX \arrow[d, "h^{\mathbb{B}}"] \\ & B \end{tikzcd}$$

A morphism of $\mathbb{S}$-algebras $(B, (-)^{\mathbb{B}})$ to $(A, (-)^{\mathbb{A}})$ can be defined as an arrow $l : B \to A$ in $\mathcal{C}$ such that the following diagram commutes for every $h : X \to B$:

$$\begin{tikzcd} SX \arrow[r, "h^{\mathbb{B}}"] \arrow[dr, "(l \cdot h)^{\mathbb{A}}"'] & B \arrow[d, "l"] \\ & A \end{tikzcd}$$

Theorem 31 (Distributive law between monads in extensive form, [MW10])

Let $\mathbb{T} = (T, \eta_T, (-)^{\mathbb{T}})$ and $\mathbb{S} = (S, \eta_S, (-)^{\mathbb{S}})$ be monads on $\mathcal{C}$. A distributive law of $\mathbb{T}$ over $\mathbb{S}$ can equivalently be given by an $\mathbb{T}$-algebra $(STA, (-)^\lambda)$ for every $A \in \mathcal{C}$ such that

- $(S\eta_T A \cdot \eta_S A)^\lambda = \eta_S TA$
- for every $f : B \to TSA$, $(f^\lambda)^{\mathbb{S}} : (STB, (-)^\lambda) \to (STA, (-)^\lambda)$ is a morphism of $\mathbb{T}$-algebras.

Unfolding the conditions for the $\mathbb{T}$-algebra and for the morphism, the following diagrams must commute for every $h : X \to STA$, $k : Y \to TX$, $f : B \to STA$ and $g : X \to STB$:

$$\begin{tikzcd} X \arrow[r, "\eta_T X"] \arrow[dr, "h"'] & TX \arrow[d, "h^\lambda"] \\ & STA \end{tikzcd} \qquad \begin{tikzcd} TY \arrow[r, "k^{\mathbb{T}}"] \arrow[dr, "(h^\lambda k)^\lambda"'] & TX \arrow[d, "h^\lambda"] \\ & STA \end{tikzcd} \qquad \begin{tikzcd} TB \arrow[r, "g^\lambda"] \arrow[dr, "((f^\lambda)^{\mathbb{S}} \cdot g)^\lambda"'] & STB \arrow[d, "(f^\lambda)^{\mathbb{S}}"] \\ & STA \end{tikzcd}$$

Proposition 32 ([MW10])

Given a distributive law λ of $\mathbb{T}$ over $\mathbb{S}$, the composite monad is given by $\mathbb{S} \circ_\lambda \mathbb{T} = (ST, S\eta_T \cdot \eta_S, \big((-)^\lambda\big)^{\mathbb{S}}\big)$

Proposition 33 ([MW10])

Let $\mathbb{S}$ and $\mathbb{T}$ be monads on $\mathcal{C}$. There is a bijection between distributive laws of $\mathbb{T}$ over $\mathbb{S}$ and monad structures $(ST, S\eta_T \cdot \eta_S, (-)^{(\mathbb{ST})})$ on ST such that:

- for every $k : Y \to TX, S(k^{\mathbb{T}}) = (\eta_S TX \cdot k)^{(\mathbb{ST})}$
- for every $m : Y \to STX$ and $h : X \to STU$:
  - $h^{(\mathbb{ST})} = (h^{(\mathbb{ST})} \cdot \eta_S TX)^{\mathbb{T}}$
  - the following diagram commutes

$$\begin{tikzcd} SY \arrow[r, "m^{\mathbb{S}}"] \arrow[dr, "(h^{(\mathbb{ST})}\cdot m)^{\mathbb{S}}"'] & STX \arrow[d, "h^{(\mathbb{ST})}"] \\ & STU \end{tikzcd}$$

1.4 2-categories

Definition 34 (2-category)

A *2-category* C consists of:

- Cellular structure:
  - A collection C_0 of 0-cells $a, b, c, \ldots$ (objects),
  - For each pair of 0-cells a, b, a collection $C_1(a, b)$ of 1-cells (morphisms) from a to b,
  - For each pair of 1-cells $f, g \in C_1(a, b)$, a collection $C_2(f, g)$ of 2-cells (2-morphisms) from f to g
- Identities:
  - A 1-cell $\mathrm{id}_a : a \to a$ for each 0-cell $a \in C_0$ (identity 1-cell),
  - A 2-cell $\mathrm{id}_f : f \Rightarrow f$ for each 1-cell $f \in C_1(a, b)$ (identity 2-cell)
- Composition:
  - A composition operation on 1-cells $g \circ f$ for every pair of composable 1-cells $f : a \to b$ and $g : b \to c$,
  - A vertical composition operation on 2-cells $\beta \circ \alpha : f \Rightarrow h$ for every pair of composable 2-cells $\alpha : f \Rightarrow g$ and $\beta : g \Rightarrow h$,
  - A horizontal composition operation on 2-cells $\beta \bullet \alpha : g \circ f \Rightarrow g' \circ f'$ for every pair of composable 2-cells $\alpha : f \Rightarrow f'$ and $\beta : g \Rightarrow g'$

  such that

  - 1-cell compositions are associative and identities are neutral
  - 2-cell vertical compositions are associative and identities are neutral
  - 2-cell horizontal compositions are associative and identities are neutral, and composition of 2-cell identities preserves 1-cell composition
  - the interchange law holds: for any composable 2-cells $\alpha, \alpha', \beta, \beta'$, we have

$$(\beta' \circ \beta) \bullet (\alpha' \circ \alpha) = (\beta' \bullet \alpha') \circ (\beta \bullet \alpha)$$

Example 35 (2-category of Categories)

The 2-category **Cat** has:

- 0-cells: categories,
- 1-cells: functors between categories,
- 2-cells: natural transformations between functors.
- pointwise composition of functors as 1-cell composition,
- pointwise composition of natural transformations as vertical composition,
- horizontal composition of natural transformations defined by the composition of functors.

Definition 36 (Adjunction in a 2-category)

Given a 2-category C, an *adjunction* between 0-cells a and b in C consists of:

- Two 1-cells $f : a \to b$ and $g : b \to a$,
- Two 2-cells $\eta : \mathrm{id}_a \Rightarrow g \circ f$ (unit) and $\varepsilon : f \circ g \Rightarrow \mathrm{id}_b$ (counit),

such that the following triangle identities hold:

$$(f \bullet \eta) \circ (\varepsilon \bullet f) = \mathrm{id}_f \qquad (\eta \bullet g) \circ (g \bullet \varepsilon) = \mathrm{id}_g$$

Definition 37 (2-functor)

Given two 2-categories C and D, a *2-functor* $F : C \to D$ consists of three mappings:

- $F : C_0 \to D_0$ that assigns to each 0-cell a in C a 0-cell Fa in D,
- $F : C_1(a, b) \to D_1(Fa, Fb)$ for each pair of 0-cells a, b in C, that assigns to each 1-cell $f : a \to b$ in C a 1-cell $Ff : Fa \to Fb$ in D,
- $F : C_2(f, g) \to D_2(Ff, Fg)$ for each pair of 1-cells $f, g : a \to b$ in C, that assigns to each 2-cell $\alpha : f \Rightarrow g$ in C a 2-cell $F\alpha : Ff \Rightarrow Fg$ in D,

such that F preserves identities and compositions of 1-cells and 2-cells.

Lemma 38

Let $F : C \to D$ be a 2-functor.

If $\langle f : a \to b, u, \eta, \varepsilon \rangle$ is an adjunction in C, then $\langle Ff : Fa \to Fb, Fu, F\eta, F\varepsilon \rangle$ is an adjunction in D.

Proof

The candidate adjunction has the right type, and a 2-functor preserves pasting diagrams and identities, so the triangle identities are preserved. □

Definition 39 (natural 2-transformation)

Let $F, G : C \to D$ be two 2-functors between 2-categories.

A *(strict) natural 2-transformation* $\begin{tikzcd} C \arrow[r, bend left, "F"] \arrow[r, bend right, "G"'] \arrow[r, phantom, "\Downarrow\Xi"] & D \end{tikzcd}$ consists of a 1-cell $\Xi_a : F_0 a \to G_0 a$ for each 0-cell a such that:

- for every 1-cell $f : a \to b$ in C:

$$\begin{tikzcd} F_0 a \arrow[r, "F_1 f"] \arrow[d, "\Xi_a"'] & F_0 b \arrow[d, "\Xi_b"] \\ G_0 a \arrow[r, "G_1 f"'] & G_0 b \end{tikzcd}$$

- for every 2-cell $\begin{tikzcd} a \arrow[r, bend left, "f"] \arrow[r, bend right, "g"'] \arrow[r, phantom, "\Downarrow\alpha"] & b \end{tikzcd}$:

$$\begin{tikzcd} & Ga \arrow[dr, bend left, "Gf"] \arrow[dr, bend right, "Gg"'] \arrow[dr, phantom, "\Downarrow G\alpha"] & \\ Fa \arrow[ur, "\Xi_a"] \arrow[dr, "Fg"'] & & Gb \\ & Fb \arrow[ur, "\Xi_b"'] & \end{tikzcd} = \begin{tikzcd} & Ga \arrow[dr, "Gf"] & \\ Fa \arrow[ur, "\Xi_a"] \arrow[dr, bend left, "Ff"] \arrow[dr, bend right, "Fg"'] \arrow[dr, phantom, "\Downarrow F\alpha"] & & Gb \\ & Fb \arrow[ur, "\Xi_b"'] & \end{tikzcd}$$

1.5 Multicategories

Definition 40 (Multicategory)

A multicategory $\mathcal{U}$ consists of:

- A class of objects $\mathrm{Ob}(\mathcal{U})$,
- For each finite (possibly empty) sequence of objects $(a_1, a_2, \ldots, a_n)$ and an object b, a set of morphisms $\mathcal{U}(a_1, a_2, \ldots, a_n; b)$,
- For each object a, an identity morphism $\mathrm{id}_a : (a) \to a$,
- A composition operation that allows for the composition of morphisms in a way that generalizes the composition in a category, satisfying appropriate associativity and identity axioms.

Example 41 (Category with Finite Products as a Multicategory)

Given a category $\mathcal{C}$ with finite products, we can construct a multicategory $\mathcal{U}$ where:

- The objects of $\mathcal{U}$ are the same as the objects of $\mathcal{C}$,
- The morphisms $\mathcal{U}(a_1, a_2, \ldots, a_n; b)$ are given by the morphisms in $\mathcal{C}$ from the product $a_1 \times a_2 \times \ldots \times a_n$ to b,
- The identities rely on the canonical isomorphism

$$\mathrm{id}_a := \prod_{i=1}^{n} a \cong a$$

- The composition is defined using the universal property of products and the composition in $\mathcal{C}$.

Example 42 (Monoidal Category as a Multicategory)

A monoidal category $(\mathcal{C}, \otimes, I)$ can be viewed as a multicategory where:

- The objects of the multicategory are the same as the objects of $\mathcal{C}$,
- The morphisms $\mathcal{U}(a_1, a_2, ..., a_n; b)$ are given by the morphisms in $\mathcal{C}$ from $a_1 \otimes a_2 \otimes ... \otimes a_n$ to b (by nesting the monoidal product)

Definition 43 (Multifunctor)

A *multi-functor* $F : \mathcal{U} \to \mathcal{V}$ between multicategories $\mathcal{U}$ and $\mathcal{V}$ consists of two mappings:

- A mapping $F : \mathrm{Ob}(\mathcal{U}) \to \mathrm{Ob}(\mathcal{V})$ that assigns to each object a in $\mathcal{U}$ an object Fa in $\mathcal{V}$,
- A mapping $F : \mathcal{U}(a_1, a_2, ..., a_n; b) \to \mathcal{V}(Fa_1, Fa_2, ..., Fa_n; Fb)$ that assigns to each morphism $f : (a_1, a_2, ..., a_n) \to b$ in $\mathcal{U}$ a morphism $Ff : (Fa_1, Fa_2, ..., Fa_n) \to Fb$ in $\mathcal{V}$,

such that F preserves:

- Identities: For each object a in $\mathcal{U}$, $F\mathrm{id}_a = \mathrm{id}_{Fa}$,
- Composition: For morphisms $f : (a_1, a_2, ..., a_n) \to b$ and $g_i : (a_{i1}, a_{i2}, ..., a_{ik_i}) \to a_i$ in $\mathcal{U}$, we have

$$F(f \circ (g_1, g_2, ..., g_n)) = Ff \circ (Fg_1, Fg_2, ..., Fg_n).$$

Definition 44 (Multi-natural transformation)

Given two multi-functors $F, G : \mathcal{U} \to \mathcal{V}$ between multicategories $\mathcal{U}$ and $\mathcal{V}$, a *multi-natural transformation* $\eta : F \to G$ consists of a family of morphisms $\{\eta_a : Fa \to Ga\}_{a \in \mathrm{Ob}(\mathcal{U})}$ such that for every morphism $f : (a_1, a_2, ..., a_n) \to b$ in $\mathcal{U}$, the following diagram commutes:

$$\begin{tikzcd}
(Fa_1, Fa_2, ..., Fa_n) \arrow[r, "Ff"] \arrow[d, "(\eta_{a_1}, \eta_{a_2}, ..., \eta_{a_n})"'] & Fb \arrow[d, "\eta_b"] \\
(Ga_1, Ga_2, ..., Ga_n) \arrow[r, "Gf"'] & Gb
\end{tikzcd}$$

Example 45 (Category of multicategories)

We have a 2-category **Multicat** where:

- 0-cells are multi-categories
- 1-cells are multi-functors
- 2-cells are multi-natural transformations

with multi-functor composition and identities:

$$\mathrm{Id}_C\, a := a \quad \mathrm{Id}_C\, f := f$$

vertical, horizontal composition of multi-natural transformations, with identities:

$$(\mathrm{id}_F)_a := \mathrm{id}_{Fa} : (a) \to a$$

Example 46 (Ordinary 2-functor)
We have a 2-functor structure $-_o$: **Multicat** → **Cat** sending:

- each multi-category $\mathcal{U}$ to its (**o**rdinary) category of morphisms:
  - $\mathrm{Ob}(\mathcal{U}_o) := \mathrm{Ob}(\mathcal{U})$
  - $\mathcal{U}_o(a, b) := \mathcal{U}(a; b)$, i.e the morphisms in $\mathcal{U}_o$ are the multi-morphisms with a length-1 source
  - identities and composition are given by identity morphisms and composition in the multi-category
- each multi-functor $F : \mathcal{U} \to \mathcal{V}$ to the functor sending:
  - $a \in \mathrm{Ob}(\mathcal{U})$ to $Fa \in \mathrm{Ob}(\mathcal{V})$
  - $f : (a) \to b$ to $F_o f := Ff : (Fa) \to Fb$
- each multi-natural transformation $\begin{tikzcd} \mathcal{U} \arrow[r, bend left, "F"] \arrow[r, bend right, "G"'] \arrow[r, phantom, "\Downarrow\alpha"] & \mathcal{V} \end{tikzcd}$ to $(\alpha_o)_a := \alpha_a$

It is 2-faithful.

1.6 Operads

Definition 47 (Operad)
An *operad* $\mathcal{O}$ consists of:

- A collection of sets $\{\mathcal{O}_n\}_{n \geq 0}$, where $\mathcal{O}_n$ is the set of n-ary operations,
- A composition function $\gamma : \mathcal{O}_n \times \mathcal{O}_{k_1} \times \cdots \times \mathcal{O}_{k_n} \to \mathcal{O}_{k_1 + \cdots + k_n}$,
- An identity element $e \in \mathcal{O}_1$,

satisfying the following axioms:

- Associativity: For all $f \in \mathcal{O}_n$, $g_i \in \mathcal{O}_{k_i}$, and $h_{ij} \in \mathcal{O}_{m_{ij}}$, we have
$$\gamma(f; g_1, \ldots, g_n) = f(g_1(h_{11}, \ldots, h_{1k_1}), \ldots, g_n(h_{n1}, \ldots, h_{nk_n})).$$
- Identity: For all $f \in \mathcal{O}_n$, we have
$$f(e, \ldots, e) = f.$$

Definition 48 (Algrebraic Theory)

An *algebraic theory* $\mathcal{T}$ consists of:

- A collection of sets $\{\mathcal{T}_n\}_{n\geq 0}$, where $\mathcal{T}_n$ is the set of n-ary operations,
- A collection of equations between terms built from these operations,

such that the operations and equations define a class of algebras (models) that satisfy the given equations.

Example 49 (Operad from an Algebraic Theory)

Given an algebraic theory $\mathcal{T}$, we can construct an operad $\mathcal{O}^{\mathcal{T}}$ where:

- $\mathcal{O}^{\mathcal{T}}_n$ consists of the n-ary operations in $\mathcal{T}$,
- The composition function γ is defined by substituting operations according to the equations of the theory,
- The identity element e corresponds to the identity operation in the theory.

Definition 50 (Renaming)

A *renaming* $\rho : n \to m$ is any function from the set of n variables to the set of m variables.

Example 51 (of renaming)

Let $\rho : 3 \to 2$ be the renaming defined by $\rho(x_1) = y_1$, $\rho(x_2) = y_2$, and $\rho(x_3) = y_1$. Then for any term $t \in \mathcal{T}_3$, we have $\rho(t) \in \mathcal{T}_2$ obtained by substituting y_1 for x_1 and x_3, and y_2 for x_2 in t.

Definition 52 (Translation)

Let $\mathcal{O}$ be an operad and $\mathcal{T}$ be a theory.

A *translation* $\mathfrak{T} : \mathcal{O} \to \mathcal{T}$ consists of an assignment $\mathfrak{T}f \in \mathcal{T}_n$ for every $f \in \mathcal{O}_n$ such that:

- the identity $\mathrm{id} : \mathcal{O}_1$ is sent the the unique variable projection: $\mathfrak{T}\mathrm{id} = x \in \mathcal{T}_1$
- the translation respects composition, for all $f \in \mathcal{O}_n$ composable with $g_i \in \mathcal{O}_{m_i}$:

$$\mathfrak{T}(f \circ (g_1, \ldots, g_n)) = \mathfrak{T}f[x_i \mapsto \mathfrak{T}g_i]_{i=1}^{n}$$

A translation is *surjective*, written $\mathfrak{T} : \mathcal{O} \twoheadrightarrow \mathcal{T}$, if for every term $t \in \mathcal{T}_n$, there exists an operator $f \in \mathcal{O}_m$ and a renaming $\rho : m \to n$ such that $t = \mathfrak{T}f[\rho]$.

Definition 53 (Model)

Let $\mathcal{O}$ be an operad, and $\mathcal{U}$ be a multicategory. A *model* of $\mathcal{O}$ in $\mathcal{U}$ is a multi-functor $\mathcal{A} : \mathcal{O} \to \mathcal{U}$ that preserves the operadic structure, i.e., it respects the composition and identity of the operad.

Definition 54

Homomorphism of Models A *homomorphism* $h : \mathcal{A} \to \mathcal{B}$ between $\mathcal{O}$-models in $\mathcal{U}$ consists of a morphism $\underline{h} : (\underline{\mathcal{A}}) \to \underline{\mathcal{B}}$ such that for all $f \in \mathcal{O}_n$,

$$\underline{h} \circ (\mathcal{A}[\![f]\!]) = \mathcal{B}[\![f]\!](\underline{h}, \ldots, \underline{h})$$

Example 55 (Category of Models)

Given an operad $\mathcal{O}$ and a multicategory $\mathcal{U}$, we can construct a category $\mathbf{Mod}(\mathcal{O},\mathcal{U})$ where:

- The objects are the models of $\mathcal{O}$ in $\mathcal{U}$,
- The morphisms are the homomorphisms between these models.

Proposition 56

Let $\mathfrak{T} : \mathcal{O} \to \mathcal{T}$ be a translation, and let $\mathcal{C}$ be a category with chosen finite product $\prod$.

There is a carrier-preserving, fully-faithful functor $\mathbf{Mod}(\mathfrak{T}, \langle \mathcal{C}, \prod \rangle) : \mathbf{Mod}(\mathcal{T},\mathcal{C}) \to \mathbf{Mod}(\mathcal{O}, \langle \mathcal{C}, \prod \rangle)$ given by:

$$\left(\mathbf{Mod}(\mathfrak{T}, \left\langle \mathcal{C}, \prod \right\rangle) A\right) [\![f]\!] := A[\![\mathfrak{T} f]\!] \qquad \mathbf{Mod}(\mathfrak{T}, \left\langle \mathcal{C}, \prod \right\rangle) h := h$$

Proposition 57

The compatibility with composition and identities of the operad follows from their preservation by $\mathfrak{T}$. Preservation of identities and composition, and fully-faithfulness are evident.

Proposition 58

We define a 2-functor $\mathbf{Mod}(\mathcal{O}, -) : \mathbf{Multicat} \to \mathbf{Cat}$ such that:

- for every multi-category $\mathcal{U}$, $\mathbf{Mod}(\mathcal{O},\mathcal{U})$ is the category of $\mathcal{O}$-models in $\mathcal{U}$.
- for every multi-functor $F : \mathcal{U} \to \mathcal{V}$, the functor $\mathbf{Mod}(\mathcal{O}, F) : \mathbf{Mod}(\mathcal{O},\mathcal{U}) \to \mathbf{Mod}(\mathcal{O},\mathcal{V})$ sends:
  - each $\mathcal{O}$-model $\mathcal{A}$ in $\mathcal{U}$ to the $\mathcal{O}$-model:
    - $\underline{F\mathcal{A}} := F\underline{\mathcal{A}}$
    - $(F\mathcal{A})[\![f]\!] : (F\underline{\mathcal{A}}, \dots, F\underline{\mathcal{A}}) \xrightarrow{F(\mathcal{A}[\![f]\!])} F\underline{\mathcal{A}}$
  - each $\mathcal{O}$-homomorphism $h : \mathcal{A} \to \mathcal{B}$ to the $\mathcal{O}$-homomorphism over $F\underline{h} : F\mathcal{A} \to F\mathcal{B}$
- for every multi-natural transformation $\begin{tikzcd} \mathcal{U} \arrow[r, bend left, "F"] \arrow[r, bend right, "G"'] \arrow[r, phantom, "\Downarrow\alpha"] & \mathcal{V} \end{tikzcd}$ we have a natural transformation

$$\begin{tikzcd} \mathbf{Mod}(\mathcal{O},\mathcal{U}) \arrow[r, bend left, "\mathbf{Mod}(\mathcal{O},F)"] \arrow[r, bend right, "\mathbf{Mod}(\mathcal{O},G)"'] \arrow[r, phantom, "\Downarrow\mathbf{Mod}(\mathcal{O},\alpha)"] & \mathbf{Mod}(\mathcal{O},\mathcal{V}) \end{tikzcd}$$ given, for every model $\mathcal{A}$, by

$$\mathbf{Mod}(\mathcal{O}, \alpha)_{\mathcal{A}} : (F\underline{\mathcal{A}}) \xrightarrow{\alpha_{\underline{\mathcal{A}}}} G\underline{\mathcal{A}}$$

1.7 Free algebras

We'll use the definitions from [All+25].

Definition 59 (Finitary signature)

A *finitary signature* $\Sigma = (\operatorname{op}\Sigma, \operatorname{arity})$ consists of a set $\operatorname{op}\Sigma$ of *operators*, and an assigment $\operatorname{arity} : \operatorname{op}\Sigma \to \mathbb{N}$ associating to each operator in $\operatorname{op}\Sigma$ its *arity*.

Definition 60 (Algebra)

An *algebra* $A = (\underline{A}, A[\![-]\!])$ over a finitary signature Σ consists of:

- A set $\underline{A}$ called the *carrier* of A,
- For each operator $f \in \mathrm{op}\,\Sigma$ of arity n, a function $A[\![f]\!] : \underline{A}^n \to \underline{A}$ called the *interpretation* of f in A.

Definition 61 (Terms)

Given a set X of variables and a finitary signature Σ, the set of *terms* $\mathrm{Term}_\Sigma(X)$ is defined inductively as follows:

- Every variable $x \in X$ is a term,
- If $f \in \mathrm{op}\,\Sigma$ is an operator of arity n and $t_1, t_2, ..., t_n$ are terms, then $f(t_1, t_2, ..., t_n)$ is a term.

Remark : The definition of algebraic theory seen before can be reformulated as a finitary signature together with a set of equations between terms built from the operators in the signature.

Definition 62 (Environment and Interpretation)

An *environment* ρ for a context X in an algebra A is a function $\rho : X \to \underline{A}$ that assigns to each variable in X an element of the carrier of A.

Given a term $t \in \mathrm{Term}_\Sigma(X)$, denoted $X \vdash t$, the algebra A gives an *interpretation* function $A[\![t]\!] : \underline{A}^X \to \underline{A}$. Given an environment $\rho : X \to \underline{A}$, we can evaluate the term t in A under the environment ρ, denoted $A[\![t]\!](\rho)$, by recursively substituting the variables in t with their corresponding values in $\underline{A}$ according to ρ and applying the interpretations of the operators as defined in A.

Definition 63 (Valid equations)

An equation $t = t'$ between terms in $\mathrm{Term}_\Sigma(X)$ is *valid* in an algebra A if for every environment $\rho : X \to \underline{A}$, we have $A[\![t]\!](\rho) = A[\![t']\!](\rho)$.

Definition 64 (Presentation and Models)

A *presentation* $\mathcal{T} = (\Sigma_\mathcal{T}, \mathcal{T}_{ax})$ consists of a finitary signature $\Sigma_\mathcal{T}$ and a set $\mathcal{T}_{ax}$ of *axioms*, which are equations between terms in $\mathrm{Term}_{\Sigma_\mathcal{T}}(X)$ for some set of variables X.

A $\mathcal{T}$-model A is a $\Sigma_\mathcal{T}$-algebra such that all equations in $\mathcal{T}_{ax}$ are valid in A.

A $\mathcal{T}$-model A over a context X is a $\mathcal{T}$-model A together with an environment $\rho : X \to \underline{A}$.

Definition 65 (Homomorphisms)

Let $A = (\underline{A}, A[\![-]\!])$ and $B = (\underline{B}, B[\![-]\!])$ be algebras over the same signature Σ.

A *homomorphism* of Σ-algebras from A to B is a function $h : \underline{A} \to \underline{B}$ such that for every operator $f \in \mathrm{op}\,\Sigma$ of arity n, we have:

$$B[\![f]\!](h(a_1), \ldots, h(a_n)) = h(A[\![f]\!](a_1, \ldots, a_n))$$

for all $a_1, \ldots, a_n \in \underline{A}$.

A *homomorphism* of $\mathcal{T}$-models is a homomorphism of $\Sigma_\mathcal{T}$-algebras between the underlying algebras of the models.

Definition 66 (Morphism)

A *morphism* $h : A \to B$ of $\mathcal{T}$-models over X is a $\mathcal{T}$-model homomorphism that makes the diagram commute:

$$\begin{tikzcd} X \arrow[r, "\rho_A"] \arrow[dr, "\rho_B"'] & \underline{A} \arrow[d, "h"] \\ & \underline{B} \end{tikzcd}$$

Definition 67 (Free model (or free algebra))

Let $\mathcal{T}$ be a presentation and X be a set of variables.

A *free $\mathcal{T}$-model* over X is a $\mathcal{T}$-model $F(X)$ such that for any $\mathcal{T}$-model A over X, there exists a unique morphism of $\mathcal{T}$-models from $F(X)$ to A.

The existence-and-uniqueness property is called the *universal property* of free $\mathcal{T}$-models over X.

Informaly:

- A signature gives defines the operators and constants (arity 0 operators).
- An algebra gives a concrete interpretation of the signature.
- A presentation defines axioms on top of the signature, for instance the axioms of a group, or of a semiring, etc.
- A model of a presentation is an algebra that satisfies the axioms of the presentation.
- A free algebra is the most general model of a presentation.

We can give an equivalent definition for free algebras using universal arrows.

Proposition 68

Let $\mathcal{T}$ be a presentation, and X be a set. Let $\mathcal{T}$-**Mod** be the category of $\mathcal{T}$-models over X. Let $U : \mathcal{T}\text{-}\mathbf{Mod} \to \mathbf{Set}$ be the forgetful functor that sends a $\mathcal{T}$-model to its underlying set.

Then, a free $\mathcal{T}$-model over X is a universal arrow from X to U.

Proof

Let $F\,X$ be a free $\mathcal{T}$-model over X. We want to show that $(F\,X, \eta_X)$ is a universal arrow from X to U, where $\eta_X : X \to U\,F\,X$ is the function that sends each variable in X to its interpretation in $F\,X$.

Let A be a $\mathcal{T}$-model over X, and let $\rho_A : X \to U\,A$ be the environment of A. Since $F\,X$ is a free $\mathcal{T}$-model over X, there exists a unique morphism of $\mathcal{T}$-models $h : F\,X \to A$ such that the following diagram commutes:

$$\begin{tikzcd} X \arrow[r, "\eta_X"] \arrow[dr, "\rho_A"'] & U\,F\,X \arrow[d, "U\,h"] \\ & U\,A \end{tikzcd}$$

This shows that $(F\,X, \eta_X)$ satisfies the universal property of being a universal arrow from X to U.

Conversely, if $(F\,X, \eta_X)$ is a universal arrow from X to U, then for any $\mathcal{T}$-model A over X, there exists a unique morphism of $\mathcal{T}$-models from $F\,X$ to A, which means that $F\,X$ is a free $\mathcal{T}$-model over X. Therefore, the two definitions are equivalent. $\square$

1.8 Multi-homomorphisms

This section will be a subset of Ohad's note. In particular, all proofs will be omitted.

In all of this section, we fix an algebraic theory $\mathcal{T}$ and $\langle \mathcal{C}, \prod \rangle$ a category with chosen finite products.

Definition 69 (Strength)

For all $k \in \mathbb{N}^*$ and $X, Y \in \mathcal{C}$, we define the *strength* (for the reader monad X^k) by

$$\mathrm{str} : X^k \times Y \xrightarrow{\langle \pi_i \times \mathrm{id} \rangle_{i=1}^k} (X \times Y)^k$$

Definition 70 (Multi-$\mathcal{T}$-homomorphism)

Let $A_1, A_2, \ldots, A_k$ and B be $\mathcal{T}$-models in $\mathcal{C}$.

A *multi-$\mathcal{T}$-homomorphism* from $A_1, A_2, \ldots, A_k$ to B is a morphism $h : \prod_{i=1}^k \underline{A_i} \to \underline{B}$ such that for every term $t \in \mathcal{T}_k$ and $1 \leqslant i_0 \leqslant n$, the following diagram commutes:

$$\begin{tikzcd}
\mathcal{A}_{i_0}^k \times \prod\limits_{\substack{1 \leqslant i \leqslant n \\ i \neq i_0}} \mathcal{A}_i \arrow[r, "\mathrm{str}"] \arrow[dd, "\mathcal{A}_{i_0}[\![t]\!] \times \mathrm{id}"'] & \left(\mathcal{A}_{i_0} \times \prod\limits_{\substack{1 \leqslant i \leqslant n \\ i \neq i_0}} \mathcal{A}_i \right)^k \arrow[r, "\cong"] & \left(\prod\limits_{i=1}^n \mathcal{A}_i \right)^k \arrow[d, "f^k"] \\
 & & \mathcal{B}^k \arrow[d, "\mathcal{B}[\![t]\!]"] \\
\mathcal{A}_{i_0} \times \prod\limits_{\substack{1 \leqslant i \leqslant n \\ i \neq i_0}} \mathcal{A}_i \arrow[r, "\cong"'] & \prod\limits_{i=1}^n \mathcal{A}_i \arrow[r, "f"'] & \mathcal{B}
\end{tikzcd}$$

▷ We can then define a multi-category structure $\mathbf{Multi}(\mathcal{T}, \mathcal{C})$ over the finite-product multi-category:

- the objects are $\mathcal{T}$-algebra
- the multi-morphisms are the multi-$\mathcal{T}$-homomorphisms
- the identities are the identity morphisms
- composition is inherited from $\langle \mathcal{C}, \prod, \rangle$

▷ We also define a multi-functor $\underline{\ } : \mathbf{Multi}(\mathcal{T}, \mathcal{C}) \to \langle \mathcal{C}, \prod \rangle$ which sends an algebra to its carrier and a multi-$\mathcal{T}$-homomorphism to its underlying morphism.

▷ We define another 2-category $\mathbf{Cat}_\times$ consisting of:

- 0-cells are $\langle \mathcal{C}, \prod \rangle$, categories with chosen finite products
- 1-cells are finite product preserving functors, not necessarily preserving the chosen products
- 2-cells are the natural transformations between these functors

Therefore, this is a sub-category of **Cat**.

▷ We define the data of a 2-functor $\mathbf{Multi}(\mathcal{T}, -) : \mathbf{Cat}_\times \to \mathbf{Multicat}$ sending:

- each category with chosen products (0-cells) $\mathcal{C}$ to the multi-category (0-cell) $\mathbf{Multi}(\mathcal{T}, \mathcal{C})$
- each finite-product preserving functor (1-cell) $H : \mathcal{C} \to \mathcal{D}$ to the following multi-functor $\mathbf{Multi}(\mathcal{T}, H) : \mathbf{Multi}(\mathcal{T}, \mathcal{C}) \to \mathbf{Multi}(\mathcal{T}, \mathcal{D})$, sending:

- each $\mathcal{T}$-algebra $\mathcal{A}$ to the $\mathcal{T}$-algebra $\mathbf{Multi}(\mathcal{T}, H) \coloneqq H\mathcal{A}$, given by:
  * its carrier $\underline{H\mathcal{A}}$ is $H\underline{\mathcal{A}}$
  * the interpretation of each $t \in \mathcal{T}_n$ is $H\mathcal{A}[\![t]\!] : (H\underline{\mathcal{A}}^n) \xrightarrow{\cong} H(\underline{\mathcal{A}}^n) \xrightarrow{\mathcal{A}[\![t]\!]} H\underline{\mathcal{A}}$
- each multi-$\mathcal{T}$-homomorphism $h : (\mathcal{A}_1, \dots, \mathcal{A}_n) \to \mathcal{B}$ to the multi-$\mathcal{T}$-homomorphism $\mathbf{Multi}(\mathcal{T}, H)h : (H\mathcal{A}_1, \dots, H\mathcal{A}_n)$ given by:

$$\mathbf{Multi}(\mathcal{T}, H)h : \prod_{i=1}^{n} H\mathcal{A}_i \xrightarrow{\cong} H\left(\prod_{i=1}^{n} \mathcal{A}_i\right) \xrightarrow{Hh} H\mathcal{B}$$

- each natural transformation $\begin{tikzcd} \mathcal{C} \arrow[r, bend left, "H"] \arrow[r, bend right, "K"'] \arrow[r, phantom, "\Downarrow\alpha"] & \mathcal{D} \end{tikzcd}$ (2-cell) to the multi-natural transformation

$\begin{tikzcd} \mathbf{Multi}(\mathcal{T},\mathcal{C}) \arrow[r, bend left, "\mathbf{Multi}(\mathcal{T},H)"] \arrow[r, bend right, "\mathbf{Multi}(\mathcal{T},K)"'] \arrow[r, phantom, "\Downarrow\mathbf{Multi}(\mathcal{T},\alpha)"] & \mathbf{Multi}(\mathcal{T},\mathcal{D}) \end{tikzcd}$, given, at an algebra $\mathcal{A} \in \mathbf{Mod}(\mathcal{T}, \mathcal{C})$, by the unary multi-$\mathcal{T}$-homomorphism corresponding to the following $\mathcal{T}$-homomorphism:

$$\mathbf{Multi}(\mathcal{T}, \alpha)'_{\mathcal{A}} : H\underline{\mathcal{A}} \xrightarrow{\alpha_{\underline{\mathcal{A}}}} K\underline{\mathcal{A}}$$

▷ We define another 2-functor $\langle - \rangle : \mathbf{Cat}_\times \to \mathbf{Multicat}$, sending:

- 0-cells: categories with specified products $\langle \mathcal{C}, \prod \rangle$ to their associated multi-category $\langle \mathcal{C}, \prod \rangle$ (cf. Example 41)
- 1-cells: finite-product preserving functors $H : \mathcal{C} \to \mathcal{D}$ to the multi-functor sending:
  - objects $a \in \mathrm{Ob}\left(\langle \mathcal{C}, \prod \rangle\right) = \mathrm{Ob}\,\mathcal{C}$ to $\langle H \rangle\, a \in \mathrm{Ob}\,\mathcal{D}$
  - multi-morphisms $f : (a_1, \dots, a_n) \to b$ to the multi-morphism:

$$\langle H \rangle\, f : \prod_{i=1}^{n} Ha_i \xrightarrow{\cong} H\left(\prod_{i=1}^{n} a_i\right) \xrightarrow{Hf} Hb$$

- 2-cells: natural transformations $\begin{tikzcd} \mathcal{C} \arrow[r, bend left, "H"] \arrow[r, bend right, "K"'] \arrow[r, phantom, "\Downarrow\alpha"] & \mathcal{D} \end{tikzcd}$ to the multi-natural transformation given for each $a \in \mathrm{Ob}(\mathcal{C})$, by:

$$\langle \alpha \rangle_a : \prod_{i=1}^{1} Ha \cong Ha \xrightarrow{\alpha_a} Ka$$

▷ Finally, we have a natural 2-transformation $\begin{tikzcd} \mathbf{Cat}_\times \arrow[r, bend left, "\mathbf{Multi}(\mathcal{T},-)"] \arrow[r, bend right, "\langle - \rangle"'] \arrow[r, phantom, "\Downarrow ="] & \mathbf{Multicat} \end{tikzcd}$ whose components, for each category $\langle \mathcal{C}, \prod \rangle$ with chosen finite products, consists of the forgetful multi-functors sending:

- each $\mathcal{T}$-algebra $\mathcal{A}$ to its carrier $\underline{\mathcal{A}}$
- each multi-$\mathcal{T}$-homomorphism $h : (\mathcal{A}_1, \dots, \mathcal{A}_n) \to \mathcal{B}$ to its underlying morphism $\underline{h} : \prod_{i=1}^{n} \underline{\mathcal{A}_i} \to \underline{\mathcal{B}}$

All of the proofs that the following definitions are indeed of the nature they were described of are in Ohad's notes.

2 Some free algebras

2.1 Semigroups

Free commutative semigroups

Let's try to construct the free algebra for the presentation of commutative semigroups. We cannot rely on the order of the terms, nor the order of applying the operations.

Let X be a set. We consider the theory of commutative semigroups **CSG**, with a signature Σ with a single operator $(+)$ of arity 2, and with the two axioms

$$a, b \vdash a + b = b + a$$
$$a, b, c \vdash a + (b + c) = (a + b) + c$$

We pose

$$SX = \{f : X \to \mathbb{N} \mid f \text{ has finite non-empty support}\}$$

For $f, g \in SX$, we pose

$$f + g = \lambda x.f\ x + g\ x \in SX$$

We define $\eta_{\mathbf{CSG}} : X \to SX$ for all $x \in X$ by

$$\eta_{\mathbf{CSG}}(x) = \lambda y. \begin{cases} 1 \text{ if } x = y \\ 0 \text{ otherwise} \end{cases}$$

We pose $F_{\mathbf{CSG}}X = (SX, (+))$ where $+$ is the operator defined above.

Proposition 71

$F_{\mathbf{CSG}}X$ is a free **CSG**-model over X.

Proof

- ▷ Let's first check that $F_{\mathbf{CSG}}X$ is indeed a **CSG**-model. We need to check that the axioms of **CSG** are valid in $F_{\mathbf{CSG}}X$. This is straightforward and comes down to the commutativity and associativity of addition in $\mathbb{N}$.

- ▷ Let A be a **CSG**-model over X with environment $\varphi_A : X \to U\,A$. We want to show that there exists a unique morphism h of **CSG**-models from $F_{\mathbf{CSG}}X$ to A such that the following diagram commutes:

$$\begin{tikzcd} X \arrow[r, "\eta_{\mathbf{CSG}}"] \arrow[dr, "\varphi_A"'] & SX \arrow[d, "h"] \\ & U\,A \end{tikzcd}$$

  with U being the forgetful functor from **CSG**-models to **Set**.

  We define $\Sigma\varphi_A : SX \to \underline{A}$ for all $f \in SX$ by

$$\Sigma\varphi_A(f) = \sum_{x \in X} f\ x \cdot \varphi_A(x)$$

  We want to show that $\Sigma\varphi_A$ is the unique morphism h of **CSG**-models from $F_{\mathbf{CSG}}X$ to A above. We need to check that $\Sigma\varphi_A$ preserves the operator $+$, and that the diagram above commutes.

– Preservation of the operator $+$: let $f, g \in SX$. We have:

$$\begin{aligned}\Sigma\varphi_A(f+g) &= \Sigma\varphi_A(\lambda x.f\ x + g\ x)\\ &= \sum_{x\in X}(f\ x + g\ x)\cdot\varphi_A(x)\\ &= \sum_{x\in X} f\ x\cdot\varphi_A(x) + \sum_{x\in X} g\ x\cdot\varphi_A(x)\\ &= \Sigma\varphi_A(f) + \Sigma\varphi_A(g)\end{aligned}$$

– Commutation of the diagram: let $x \in X$. We have:

$$\begin{aligned}(\Sigma\varphi_A)(\eta_{\mathbf{CSG}}(x)) &= \Sigma\varphi_A\left(\lambda y.\begin{cases}1 \text{ if } x = y\\ 0 \text{ otherwise}\end{cases}\right)\\ &= \sum_{y\in X}\begin{cases}1\cdot\varphi_A(x) & \text{if } x = y\\ 0 & \text{otherwise}\end{cases}\\ &= \varphi_A(x)\end{aligned}$$

Therefore, the diagram commutes.

– Uniqueness: let $h : F_{\mathbf{CSG}}X \to A$ be a morphism of **CSG**-models such that the diagram above commutes. We want to show that $h = \Sigma\varphi_A$. Let $f \in SX$. We have:

$$\begin{aligned}h(f) &= h\left(\sum_{x\in X} f\ x\cdot\eta_{\mathbf{CSG}}(x)\right)\\ &= \sum_{x\in X} f\ x\cdot h(\eta_{\mathbf{CSG}}(x))\\ &= \sum_{x\in X} f\ x\cdot\varphi_A(x) && \text{since the diagram commutes}\\ &= \Sigma\varphi_A(f)\end{aligned}$$

$\boxed{\text{Therefore, we proved that } F_{\mathbf{CSG}}X \text{ is a free } \mathbf{CSG}\text{-model over } X.}$

□

Finally we'll define a multiplication $\mu_{\mathbf{CSG}}$ so that we have a monad structure $(S, \eta_{\mathbf{CSG}}, \mu_{\mathbf{CSG}})$ on **Set**. For all $f \in SSX$, we pose

$$\mu_{\mathbf{CSG}}(f) = \lambda x.\sum_{g\in SX} f\ g\cdot g\ x$$

Proposition 72

$(S, \eta_{\mathbf{CSG}}, \mu_{\mathbf{CSG}})$ is a monad on **Set**.

Proof

We need to check the monad axioms:

- Left unit: let $f \in SX$. We have:

$$\begin{aligned}\mu_{\mathbf{CSG}}(\eta_{\mathbf{CSG}}S(f)) &= \mu_{\mathbf{CSG}}\left(\lambda g.\begin{cases}1 \text{ if } g = f\\ 0 \text{ otherwise}\end{cases}\right)\\ &= \lambda x.\sum_{g\in SX}\begin{cases}1\cdot g\ x & \text{if } g = f\\ 0 & \text{otherwise}\end{cases}\\ &= \lambda x.f\ x\\ &= f\end{aligned}$$

- Right unit: let $f \in SX$. We have:

$$\begin{aligned}
\mu_{\mathbf{CSG}}(S\eta_{\mathbf{CSG}}(f)) &= \mu_{\mathbf{CSG}}\left(\lambda g.\begin{cases} f\ x & \text{if } g = \eta_{\mathbf{CSG}}(x) \text{ for some } x \in X \\ 0 & \text{otherwise}\end{cases}\right) \\
&= \lambda x.\sum_{g \in SX}\begin{cases} f\ x \cdot g\ x & \text{if } g = \eta_{\mathbf{CSG}}(x) \text{ for some } x \in X \\ 0 & \text{otherwise}\end{cases} \\
&= \lambda x.f(x) \cdot \eta_{\mathbf{CSG}}(x)(x) \\
&= \lambda x.f\ x \\
&= f
\end{aligned}$$

- Associativity: let $f \in SSSX$. We have:

$$\begin{aligned}
\mu_{\mathbf{CSG}}(S\mu_{\mathbf{CSG}}(f)) &= \mu_{\mathbf{CSG}}\left(\lambda g.\sum_{h \in SSX} f\ h \cdot h\ g\right) \\
&= \lambda x.\sum_{g \in SX}\left(\sum_{h \in SSX} f\ h \cdot h\ g\right) \cdot g\ x \\
&= \lambda x.\sum_{h \in SSX} f\ h \cdot \left(\sum_{g \in SX} h\ g \cdot g\ x\right) \\
&= \lambda x.\sum_{h \in SSX} f\ h \cdot \mu_{\mathbf{CSG}}(h) \\
&= \mu_{\mathbf{CSG}}(\mu_{\mathbf{CSG}}S(f))
\end{aligned}$$

□

Free semigroups

Compared to the free commutative semigroup, we can rely on the order of the terms. We'll use lists to represent the terms of the free semigroup.

Let X be a set. We consider the theory of semigroups **SG**, with a signature Σ with a single operator $(\cdot)$ of arity 2, and with the axiom

$$a, b, c \vdash a \cdot (b \cdot c) = (a \cdot b) \cdot c$$

We pose

$$TX = \{l \in X^* \mid l \text{ is finite and non-empty}\}$$

For $f, g \in TX$, we pose

$$f \cdot g = f +\!\!+\ g$$

where $+\!\!+$ is the concatenation of lists.

We define $\eta_{\mathbf{SG}} : X \to TX$ for all $x \in X$ by

$$\eta_{\mathbf{SG}}(x) = [x]$$

We pose $F_{\mathbf{SG}}X = (TX, (\cdot))$ where $\cdot$ is the operator defined above.

Proposition 73

$F_{\mathbf{SG}}X$ is a free **SG**-model over X.

Proof

▷ $F_{\mathbf{SG}}$ is a **SG**-model: the associativity of $\cdot$ in $F_{\mathbf{SG}}$ comes from the associativity of list concatenation.

▷ Let A be a **SG**-model over X with environment $\varphi_A : X \to U\,A$.

We define $\Sigma\varphi_A : TX \to \underline{A}$ for all $l \in TX$ by

$$\Sigma\varphi_A(l) = \prod_{x \in l} \varphi_A(x)$$

where the product[a] is taken in the order of the list l, which is well defined because the list is finite, non-empty, and the product is associative.

Let's show that $\Sigma\varphi_A$ is the unique morphism h of **SG**-models from $F_{\mathbf{SG}}X$ to A such that the following diagram commutes:

$$\begin{tikzcd} X \arrow[r, "\eta_{\mathbf{SG}}"] \arrow[dr, "\varphi_A"'] & TX \arrow[d, "h"] \\ & U\,A \end{tikzcd}$$

– Preservation of the operator $\cdot$: let $f, g \in TX$. We have:

$$\begin{aligned} \Sigma\varphi_A(f \cdot g) &= \Sigma\varphi_A(f +\!\!+\, g) \\ &= \prod_{x \in f +\!\!+ g} \varphi_A(x) \\ &= \prod_{x \in f} \varphi_A(x) \cdot \prod_{x \in g} \varphi_A(x) && \text{by associativity of the product} \\ &= \Sigma\varphi_A(f) \cdot \Sigma\varphi_A(g) \end{aligned}$$

– Commutation of the diagram: let $x \in X$. We have:

$$\begin{aligned} (\Sigma\varphi_A)(\eta_{\mathbf{SG}}(x)) &= \Sigma\varphi_A([x]) \\ &= \varphi_A(x) \end{aligned}$$

– Uniqueness: let $h : F_{\mathbf{SG}}X \to A$ be a morphism of **SG**-models such that the diagram above commutes. We want to show that $h = \Sigma\varphi_A$. Let $l \in TX$. We have:

$$\begin{aligned} h(l) &= h\left(\prod_{x \in l} \eta_{\mathbf{SG}}(x)\right) \\ &= \prod_{x \in l} h(\eta_{\mathbf{SG}}(x)) && \text{since } h \text{ is a morphism} \\ &= \prod_{x \in l} \varphi_A(x) && \text{since the diagram commutes} \\ &= \Sigma\varphi_A(l) \end{aligned}$$

$\boxed{\text{Therefore we proved that } F_{\mathbf{SG}}X \text{ is a free } \mathbf{SG}\text{-model over } X.}$

□

[a] product with the operator $\cdot$

We'll also define a multiplication $\mu_{\mathbf{SG}}$ so that we have a monad structure $(T, \eta_{\mathbf{SG}}, \mu_{\mathbf{SG}})$ on **Set**. For all $l \in TTX$, we pose

$$\mu_{\mathbf{SG}}(l) = +\!\!+_{x \in l}\, x$$

where the concatenation is taken in the order of the list l, which is well defined because the list l is finite, non-empty, and the concatenation is associative.

Proposition 74

$(T, \eta_{\mathbf{SG}}, \mu_{\mathbf{SG}})$ is a monad on **Set**.

Proof

We need to check the monad axioms:

- Left unit: let $l \in TX$. We have:

$$\begin{aligned}\mu_{\mathbf{SG}}(\eta_{\mathbf{SG}}T(l)) &= \mu_{\mathbf{SG}}([l]) \\ &= +\!\!+_{x\in[l]} x \\ &= l\end{aligned}$$

- Right unit: let $l \in TX$. We have:

$$\begin{aligned}\mu_{\mathbf{SG}}(T\eta_{\mathbf{SG}}(l)) &= \mu_{\mathbf{SG}}\left([[x_i] \mid x_i \in l]\right) \\ &= +\!\!+_{x\in[[x_i]|x_i\in l]} x \\ &= +\!\!+_{x_i\in l}[x_i] \\ &= l\end{aligned}$$

- Associativity: let $l \in TTTX$. We have:

$$\begin{aligned}\mu_{\mathbf{SG}}(T\mu_{\mathbf{SG}}(l)) &= \mu_{\mathbf{SG}}\left([+\!\!+_{x\in l_i} x \mid l_i \in l]\right) \\ &= +\!\!+_{x\in[+\!\!+_{x\in l_i} x|l_i\in l]} x \\ &= +\!\!+_{l_i\in l} +\!\!+_{x\in l_i} x \\ &= +\!\!+_{x\in +\!\!+_{l_j\in l} l_j} x \\ &= +\!\!+_{x\in l} x \\ &= \mu_{\mathbf{SG}}(\mu_{\mathbf{SG}}T(l))\end{aligned}$$

□

2.2 A first combination

We will call $\mathbf{SG}+\mathbf{CSG}$ the distributive combination of **SG** and **CSG**.

The signature of $\mathbf{SG}+\mathbf{CSG}$ has two operators: $(\cdot)$ of arity 2, and $(+)$ of arity 2, and the axioms are the union of the axioms of **SG** and **CSG** :

$$\begin{aligned}a,b,c &\vdash a\cdot(b\cdot c) = (a\cdot b)\cdot c \\ a,b &\vdash a+b = b+a \\ a,b,c &\vdash a+(b+c) = (a+b)+c\end{aligned}$$

We also have the distributivity axioms:

$$\begin{aligned}a,b,c &\vdash a\cdot(b+c) = a\cdot b + a\cdot c && \text{left distributivity} \\ a,b,c &\vdash (a+b)\cdot c = a\cdot c + b\cdot c && \text{right distributivity}\end{aligned}$$

We'll try to combine the two previous examples to hopefully get the free algebra for the presentation.

The carrier set of our candidate for the free $\mathbf{SG}+\mathbf{CSG}$-model over X is STX.

We will keep the same definition for the operator $+$ as in the free commutative semigroup, but we'll need to change the definition of the operator $\cdot$ to take into account the fact that we are now working with STX instead of TX.

If we write $(p : n) \in f$ when $f(p) = n$, then what we want is

$$f \cdot g = \sum_{(p:n)\in f} \sum_{(q:m)\in g} (p \cdot_{\mathbf{SG}} q : n \cdot m)$$

Formaly, this can be defined for all $f, g \in STX$ as

$$f \cdot g = \lambda l : TX. \sum_{l = l_1 \cdot_{\mathbf{SG}} l_2} f(l_1) \cdot g(l_2)$$

We define $\eta_{\mathbf{SG+CSG}} : X \to STX = \eta_{\mathbf{CSG}} \circ \eta_{\mathbf{SG}}$. Let's first prove that the following diagram commutes:

$$\begin{tikzcd} X \arrow[r, "\eta_{\mathbf{SG}}"] \arrow[dr, "\eta_{\mathbf{CSG}}"'] & TX \arrow[r, "\eta_{\mathbf{CSG}}T"] & STX \\ & SX \arrow[ur, "S\eta_{\mathbf{SG}}"'] & \end{tikzcd}$$

Let $x \in X$. We have:

$$\begin{aligned} (S\eta_{\mathbf{SG}})(\eta_{\mathbf{CSG}}(x)) &= S\eta_{\mathbf{SG}}\left(\lambda y. \begin{cases} 1 \text{ if } x = y \\ 0 \text{ otherwise} \end{cases}\right) \\ &= \lambda y. \begin{cases} 1 & \text{if } \eta_{\mathbf{SG}}(x) = y \\ 0 & \text{otherwise} \end{cases} \\ &= \eta_{\mathbf{CSG}}(\eta_{\mathbf{SG}}(x)) \end{aligned}$$

We pose $F_{\mathbf{SG+CSG}}X = (STX, (\cdot), (+))$, where $(\cdot)$ and $(+)$ are defined above.

Proposition 75

$F_{\mathbf{SG+CSG}}X$ is a free $\mathbf{SG} + \mathbf{CSG}$-model over X.

Proof

▷ Let's first check that $F_{\mathbf{SG+CSG}}X$ is indeed a $\mathbf{SG} + \mathbf{CSG}$-model.

The associativity and commutativity of $(+)$ was already proven for $F_{\mathbf{CSG}}X$.

For the associativity of $(\cdot)$: let $f, g, h \in STX$. We have:

$$\begin{aligned} (f \cdot g) \cdot h &= \lambda l : TX. \sum_{l = l_1 \cdot_{\mathbf{SG}} l_2} (f \cdot g)(l_1) \cdot h(l_2) \\ &= \lambda l : TX. \sum_{l = l_1 \cdot_{\mathbf{SG}} l_2} \sum_{l_1 = l_{11} \cdot_{\mathbf{SG}} l_{12}} f(l_{11}) \cdot g(l_{12}) \cdot h(l_2) \\ &= \lambda l : TX. \sum_{l = l_{11} \cdot_{\mathbf{SG}} (l_{12} \cdot_{\mathbf{SG}} l_2)} f(l_{11}) \cdot g(l_{12}) \cdot h(l_2) && \text{by associativity of } \cdot_{\mathbf{SG}} \\ &= \lambda l : TX. \sum_{l = l_{11} \cdot_{\mathbf{SG}} l_3} \sum_{l_3 = l_{12} \cdot_{\mathbf{SG}} l_2} f(l_{11}) \cdot g(l_{12}) \cdot h(l_2) \\ &= \lambda l : TX. \sum_{l = l_{11} \cdot_{\mathbf{SG}} l_3} f(l_{11}) \cdot (g \cdot h)(l_3) \\ &= f \cdot (g \cdot h) \end{aligned}$$

The distributivity of $(\cdot)$ over $(+)$ comes from the distributivity of the product in $\mathbb{N}$ over addition, and from the definition of $(+)$.

Therefore, $F_{\mathbf{SG+CSG}}X$ is a $\mathbf{SG} + \mathbf{CSG}$-model.

▷ Let A be a $\mathbf{SG} + \mathbf{CSG}$-model over X with environment $\varphi_A : X \to UA$. We want to show that there exists a unique morphism h of $\mathbf{SG} + \mathbf{CSG}$-models from $F_{\mathbf{SG+CSG}}X$ to A such that the following diagram commutes:

$$\begin{tikzcd} X \arrow[r, "\eta_{\mathbf{SG}+\mathbf{CSG}}"] \arrow[dr, "\varphi_A"'] & STX \arrow[d, "h"] \\ & UA \end{tikzcd}$$

We define $\Sigma\varphi_A : STX \to \underline{A}$ for all $f \in STX$ by

$$\Sigma\varphi_A(f) = \sum_{l \in TX} f\ l \cdot \prod_{x \in l} \varphi_A(x)$$

Let's show that $\Sigma\varphi_A$ is the unique morphism h of $\mathbf{SG} + \mathbf{CSG}$-models from $F_{\mathbf{SG}+\mathbf{CSG}}X$ to A above. We need to check that $\Sigma\varphi_A$ preserves the operators $(\cdot)$ and $(+)$, and that the diagram above commutes.

- Preservation of the operator $(\cdot)$: let $f, g \in STX$. We have:

$$\begin{aligned}
\Sigma\varphi_A(f \cdot g) &= \Sigma\varphi_A(\lambda l : TX. \sum_{l = l_1 \cdot_{\mathbf{SG}} l_2} f(l_1) \cdot g(l_2)) \\
&= \sum_{l \in TX} \left(\sum_{l = l_1 \cdot_{\mathbf{SG}} l_2} f(l_1) \cdot g(l_2) \right) \cdot \prod_{x \in l} \varphi_A(x) \\
&= \sum_{l_1, l_2 \in TX} f(l_1) \cdot g(l_2) \cdot \prod_{x \in l_1 \cdot_{\mathbf{SG}} l_2} \varphi_A(x) \\
&= \sum_{l_1, l_2 \in TX} f(l_1) \cdot g(l_2) \cdot \prod_{x \in l_1} \varphi_A(x) \cdot \prod_{x \in l_2} \varphi_A(x) \qquad \text{by associativity of the product} \\
&= \sum_{l_1 \in TX} f(l_1) \cdot \prod_{x \in l_1} \varphi_A(x) \cdot \sum_{l_2 \in TX} g(l_2) \cdot \prod_{x \in l_2} \varphi_A(x) \\
&= \Sigma\varphi_A(f) \cdot \Sigma\varphi_A(g)
\end{aligned}$$

- Preservation of the operator $(+)$: let $f, g \in STX$. We have:

$$\begin{aligned}
\Sigma\varphi_A(f + g) &= \Sigma\varphi_A(\lambda l : TX. f\ l + g\ l) \\
&= \sum_{l \in TX} (f\ l + g\ l) \cdot \prod_{x \in l} \varphi_A(x) \\
&= \sum_{l \in TX} f\ l \cdot \prod_{x \in l} \varphi_A(x) + \sum_{l \in TX} g\ l \cdot \prod_{x \in l} \varphi_A(x) \\
&= \Sigma\varphi_A(f) + \Sigma\varphi_A(g)
\end{aligned}$$

- Commutation of the diagram: let $x \in X$. We have:

$$\begin{aligned}
(\Sigma\varphi_A)(\eta_{\mathbf{SG}+\mathbf{CSG}}(x)) &= \Sigma\varphi_A(\eta_{\mathbf{CSG}}(\eta_{\mathbf{SG}}(x))) \\
&= \Sigma\varphi_A\left(\lambda y. \begin{cases} 1 & \text{if } \eta_{\mathbf{SG}}(x) = y \\ 0 & \text{otherwise} \end{cases} \right) \\
&= \sum_{y \in TX} \begin{cases} 1 \cdot \prod_{x' \in y} \varphi_A(x') & \text{if } \eta_{\mathbf{SG}}(x) = y \\ 0 & \text{otherwise} \end{cases} \\
&= \prod_{x' \in \eta_{\mathbf{SG}}(x)} \varphi_A(x') \\
&= \varphi_A(x)
\end{aligned}$$

– Uniqueness: let $h : F_{\mathbf{SG}+\mathbf{CSG}}X \to A$ be a morphism of $\mathbf{SG}+\mathbf{CSG}$-models such that the diagram above commutes. We want to show that $h = \Sigma\varphi_A$. Let $f \in STX$. We have:

$$\begin{aligned}
h(f) &= h\left(\sum_{l\in TX} f\ l \cdot \prod_{x\in l} \eta_{\mathbf{SG}}(x)\right) \\
&= \sum_{l\in TX} f\ l \cdot \prod_{x\in l} h(\eta_{\mathbf{SG}}(x)) && \text{since } h \text{ is a morphism} \\
&= \sum_{l\in TX} f\ l \cdot \prod_{x\in l} \varphi_A(x) && \text{since the diagram commutes} \\
&= \Sigma\varphi_A(f)
\end{aligned}$$

$\boxed{\text{Therefore } F_{\mathbf{SG}+\mathbf{CSG}}X \text{ is a free } \mathbf{SG}+\mathbf{CSG}\text{-model over } X.}$

□

We also define a distributive law $\lambda : TSX \to STX$ for all $l \in TSX$ by

$$\lambda([f_1,\dots,f_n]) = \sum_{(x_1:n_1)\in f_1} \cdots \sum_{(x_n:n_n)\in f_n} \left([x_1,\cdots,x_n] : \prod_{i=1}^{n} n_i\right)$$

Formaly, this can be defined for all $l \in TSX$ as

$$\lambda(l) = \lambda x : TX. \begin{cases} 0 & \text{if } |x| \neq |l| \\ \prod_{(y,f)\in x*l} f(y) & \text{otherwise} \end{cases}$$

where $x * l$ is the elementwise pairing of the list x and the list l, which is well defined because x and l have the same length.

Proposition 76

λ is a distributive law of the monad T over the monad S.

Proof

We need to check the following axioms:

- $\lambda \circ T\eta_S = \eta_S T$: let $l \in TX$. We have:

$$\begin{aligned}
(\lambda \circ T\eta_S)(l) &= \lambda(T\eta_S(l)) \\
&= \lambda\left([\eta_S(x) \mid x \in l]\right) \\
&= \lambda x : TX. \begin{cases} 0 & \text{if } |x| \neq |l| \\ \prod_{(y,f)\in x*[\eta_S(x)|x\in l]} f(y) & \text{otherwise} \end{cases} \\
&= \lambda x : TX. \begin{cases} 0 & \text{if } |x| \neq |l| \\ \prod_{(y,z)\in x*l} \eta_S(z)(y) & \text{otherwise} \end{cases} \\
&= \eta_S T(l)
\end{aligned}$$

- $\lambda \circ \eta_T S = S\eta_T$: let $f \in SX$. We have:

$$\begin{aligned}(\lambda \circ \eta_T S)(f) &= \lambda(\eta_T S(f)) \\ &= \lambda([f]) \\ &= \lambda x : TX. \begin{cases} 0 & \text{if } |x| \neq 1 \\ \prod_{(y,g)\in x * [f]} g(y) & \text{otherwise} \end{cases} \\ &= S\eta_T(f)\end{aligned}$$

- $\lambda \circ T\mu_S = \mu_S T \circ S\lambda \circ \lambda S$: let $l = [f_1, \dots, f_n] \in TSSX$. We have:

$$\begin{aligned}(\lambda \circ T\mu_S)(l) &= \lambda(T\mu_S(l)) \\ &= \lambda\left([\mu_S(f_i) \mid i = 1, \dots, n]\right) \\ &= \lambda x : TX. \begin{cases} \prod_{i=1}^{n} \mu_S(f_i)(x_i) & x = [x_1, \dots, x_n] \\ 0 & \text{otherwise} \end{cases} \\ &= \lambda x : TX. \begin{cases} \prod_{i=1}^{n} \sum_{g\in SX} f_i(g) \cdot g(x_i) & x = [x_1, \dots, x_n] \\ 0 & \text{otherwise} \end{cases} \\ &= \lambda x : TX. \begin{cases} \sum_{g_1,\dots,g_n \in SX} \prod_{i=1}^{n} f_i(g_i) \cdot g_i(x_i) & x = [x_1, \dots, x_m] \\ 0 & \text{otherwise} \end{cases}\end{aligned}$$

$$\begin{aligned}(\mu_S T \circ S\lambda \circ \lambda S)(l) &= \mu_S T(S\lambda(\lambda S(l))) \\ &= \mu_S T\left(S\lambda\left(\lambda y : TSX. \begin{cases} \prod_{i=1}^{n} \sum_{g\in SX} f_i(g) \cdot g(y_i) & y = [y_1, \dots, y_n] \\ 0 & \text{otherwise} \end{cases}\right)\right) \\ &= \mu_S T\left(\lambda h : STX. \sum_{g_1,\dots,g_n \in SX \wedge \lambda([g_1,\dots,g_n]) = h} \prod_{i=1}^{n} f_i(g_i) \cdot g_i(x_i)\right) \\ &= \lambda x : TX. \sum_{g_1,\dots,g_n \in SX} \left(\prod_{i=1}^{n} f_i(g_i)\right) \cdot \lambda([g_1, \dots, g_n])(x) \\ &= \lambda x : TX. \begin{cases} \sum_{g_1,\dots,g_n \in SX} \left(\prod_{i=1}^{n} f_i(g_i)\right) \cdot \left(\prod_{i=1}^{n} g_i(x_i)\right) & x = [x_1, \dots, x_n] \\ 0 & \text{otherwise} \end{cases}\end{aligned}$$

Therefore, we have $(\lambda \circ T\mu_S)(l) = (\mu_S T \circ S\lambda \circ \lambda S)(l)$.

- $\lambda \circ \mu_T S = S\mu_T \circ \lambda T \circ T\lambda$: let $l = [[f_{11}, \dots, f_{1m_1}], \dots, [f_{n1}, \dots, f_{nm_n}]] \in TTSX$. We have:

$$\begin{aligned}(\lambda \circ \mu_T S)(l) &= \lambda(\mu_T S(l)) \\ &= \lambda(+\!\!+_{i=1}^{n} [f_{i1}, \dots, f_{im_i}]) \\ &= \lambda x : TX. \begin{cases} \prod_{i=1}^{n} \prod_{j=1}^{m_i} f_{ij}(x_{ij}) & x = +\!\!+_{i=1}^{n} [x_{i1}, \dots, x_{im_i}] \\ 0 & \text{otherwise} \end{cases}\end{aligned}$$

$$
\begin{aligned}
&(S\mu_T \circ \lambda T \circ T\lambda)(l)\\
&= S\mu_T(\lambda T(T\lambda(l)))\\
&= S\mu_T(\lambda T([\lambda([f_{11},\dots,f_{1m_1}]),\dots,\lambda([f_{n1},\dots,f_{nm_n}])]))\\
&= S\mu_T\left(\lambda T\left(\left[\lambda y : TX.\begin{cases}\prod_{j=1}^{m_1} f_{1j}(y_j) & y=[y_1,\dots,y_{m_1}]\\ 0 & \text{otherwise}\end{cases},\dots,\lambda y : TX.\begin{cases}\prod_{j=1}^{m_n} f_{nj}(y_j) & y=[y_1,\dots,y_{m_n}]\\ 0 & \text{otherwise}\end{cases}\right]\right)\right)\\
&= S\mu_T\left(\lambda h : STX.\sum_{i=1}^{n}\sum_{y_1,\dots,y_{m_i}\in TX}\begin{cases}\prod_{j=1}^{m_i} f_{ij}(y_j) & h=\lambda y : TX.\begin{cases}\prod_{j=1}^{m_i} f_{ij}(y_j) & y=[y_1,\dots,y_{m_i}]\\ 0 & \text{otherwise}\end{cases}\\ 0 & \text{otherwise}\end{cases}\right)\\
&= \lambda x : TX.\begin{cases}\sum_{i=1}^{n}\sum_{y_1,\dots,y_{m_i}\in TX}\begin{cases}\prod_{j=1}^{m_i} f_{ij}(y_j) & y_i=[y_{i1},\dots,y_{im_i}]\\ 0 & \text{otherwise}\end{cases} & x = +\!\!+_{i=1}^{n}[y_1,\dots,y_{m_i}]\\ 0 & \text{otherwise}\end{cases}\\
&= \lambda x : TX.\begin{cases}\prod_{i=1}^{n}\prod_{j=1}^{m_i} f_{ij}(x_{ij}) & x = +\!\!+_{i=1}^{n}[x_{i1},\dots,x_{im_i}]\\ 0 & \text{otherwise}\end{cases}
\end{aligned}
$$

Therefore, we have $(\lambda \circ \mu_T S)(l) = (S\mu_T \circ \lambda T \circ T\lambda)(l)$.

□

Corollary 77

$(ST, \eta_{\mathbf{SG+CSG}}, \mu_{\mathbf{SG+CSG}})$ is a monad on **Set**, where

$$\mu_{\mathbf{SG+CSG}} : STST \xrightarrow{S\lambda T} SSTT \xrightarrow{\mu_S\mu_T} ST$$

Another approach to get a monad structure on ST is to use monads in extension form.

[MW10] shows[1] that we can define an **SG**-algebra $(STX, (-)^{\lambda})$ where for all $f : X \to STY$, we have $f^{\lambda} : TX \to STY$ defined by $f^{\lambda}([x_1,\dots,x_n]) = \lambda([f(x_1),\dots,f(x_n)])$. It is then showed that this **SG**-algebra gives a distributive law of the monad T over the monad S, and therefore a monad structure on ST.

In extensive form, the monad is $(ST, \eta_{\mathbf{SG+CSG}}, (-)^{(ST)})$ where for all $f : X \to STY$, we have $f^{(ST)} : STX \to STY$ defined by $f^{(ST)} = (f^{\lambda})^{T}$.
We also have that $\lambda = (S\eta_T)^{\lambda}$ as expected.

We now may want to see whether the monad we got from the distributive law is the same as the one we get from the free $\mathbf{SG} + \mathbf{CSG}$-model construction.
Informally, we want to check whether the following diagram commutes (also see Definition 83):

$$
\begin{tikzcd}
\mathbf{SG}, \mathbf{CSG} \arrow[r, "\text{combining theory}"] \arrow[d, "\text{adding monad structure}"'] \arrow[dr, phantom, "\overset{?}{=}"] & \mathbf{SG} \lhd \mathbf{CSG} \arrow[d, "\text{adding monad structure}"] \\
\mathbf{Mon}(\mathbf{SG}), \mathbf{Mon}(\mathbf{CSG}) \arrow[r, "\text{distributive law}"'] & \mathbf{Mon}(\mathbf{SG} \lhd \mathbf{CSG})
\end{tikzcd}
$$

[1] example 8.5

The classical monadic structure on **SG** + **CSG** comes from the adjunction between **Set** and **SG** + **CSG**-models given by the free **SG** + **CSG**-model construction.

Lemma 78
$F_{\mathbf{SG+CSG}} \dashv U$ is an adjunction, where $U : \mathbf{SG+CSG\text{-}Mod} \to \mathbf{Set}$ is the forgetful functor.

Proof
We need to show that for all $X \in \mathbf{Set}$ and all **SG** + **CSG**-model A, there is a natural bijection between **SG** + **CSG**-model morphisms from $F_{\mathbf{SG+CSG}}X$ to A and functions from X to $U\,A$.

Let $X \in \mathbf{Set}$ and let $A = (\underline{A}, \cdot_A, +_A)$ be a **SG** + **CSG**-model. We have already shown that there is a unique morphism of **SG** + **CSG**-models from $F_{\mathbf{SG+CSG}}X$ to A such that the diagram below commutes:

$$\begin{tikzcd} X \arrow[r, "\eta_{\mathbf{SG+CSG}}"] \arrow[dr, "\varphi_A"'] & STX \arrow[d, "\Sigma\varphi_A"] \\ & \underline{A} \end{tikzcd}$$

Therefore, we have a function $\Phi_{X,A}$ from **SG** + **CSG**-model morphisms from $F_{\mathbf{SG+CSG}}X$ to A to functions from X to $U\,A$ defined by $\Phi_{X,A}(h) = h \circ \eta_{\mathbf{SG+CSG}}$.

Conversely, we have a function $\Psi_{X,A}$ from functions from X to $U\,A$ to **SG** + **CSG**-model morphisms from $F_{\mathbf{SG+CSG}}X$ to A defined by $\Psi_{X,A}(\varphi_A) = \Sigma\varphi_A$.

We need to show that $\Phi_{X,A}$ and $\Psi_{X,A}$ are inverses of each other.
Let $h : F_{\mathbf{SG+CSG}}X \to A$ be a morphism of **SG** + **CSG**-models. We have:

$$\begin{aligned}(\Psi_{X,A} \circ \Phi_{X,A})(h) &= \Psi_{X,A}(h \circ \eta_{\mathbf{SG+CSG}}) \\ &= \Sigma(h \circ \eta_{\mathbf{SG+CSG}}) \\ &= h\end{aligned}$$

Let $\varphi_A : X \to U\,A$ be a function. We have:

$$\begin{aligned}(\Phi_{X,A} \circ \Psi_{X,A})(\varphi_A) &= \Phi_{X,A}(\Sigma\varphi_A) \\ &= \Sigma\varphi_A \circ \eta_{\mathbf{SG+CSG}} \\ &= \varphi_A\end{aligned}$$

Therefore, $\Phi_{X,A}$ and $\Psi_{X,A}$ are inverses of each other, and we have a natural bijection between **SG** + **CSG**-model morphisms from $F_{\mathbf{SG+CSG}}X$ to A and functions from X to $U\,A$. □

This adjunction gives us a monad structure on ST.

Theorem 79
The monad structure on ST given by the adjunction $F_{\mathbf{SG+CSG}} \dashv U$ is the same as the one given by the distributive law λ.

Proof
We'll consider the extensive form of the monads.

▷ **Carrier sets.**

Let $X \in \mathbf{Set}$. $UF_{\mathbf{SG+CSG}}X$ is the carrier set of the free **SG** + **CSG**-model over X, which is STX. The carrier set of the monad given by the distributive law is also STX. Therefore, the carrier sets are the same.

▷ **Unit.**

The unit is the same for both monads, sending $x \in X$ to the sum of the singleton list $[x]$ in STX.

▷ **Extension.**

Let $f : X \to STY$. Since $F_{\mathbf{SG+CSG}}Y$ is a model of $\mathbf{SG} + \mathbf{CSG}$, freeness of $F_{\mathbf{SG+CSG}}X$ gives us a unique morphism of $\mathbf{SG} + \mathbf{CSG}$-models from $F_{\mathbf{SG+CSG}}X$ to $F_{\mathbf{SG+CSG}}Y$ such that the diagram below commutes:

$$\begin{tikzcd} X \arrow[r, "\eta_{\mathbf{SG+CSG}}"] \arrow[dr, "f"'] & STX \arrow[d, "\Sigma f"] \\ & STY \end{tikzcd}$$

Therefore, the extension of f in the monad given by the adjunction is Σf. The extension of f in the monad given by the distributive law is $f^{(ST)} = (f^\lambda)^T$. We need to show that $\Sigma f = (f^\lambda)^T$.

Let $f : X \to STY$. Suppose that

$$f(x_k) = \sum_j u_{k,j}$$

for some $u_{k,j} \in TY$. Then

$$\Sigma f([x_1, \ldots, x_n]) = \sum_{j_1,\ldots,j_n} u_{1,j_1} \cdots u_{n,j_n}$$

On the other hand, we have:

$$\begin{aligned} f^\lambda([x_1, \ldots, x_n]) &= \lambda([f(x_1), \ldots, f(x_n)]) \\ &= \lambda([\sum_{j_1} u_{1,j_1}, \ldots, \sum_{j_n} u_{n,j_n}]) \\ &= \sum_{j_1,\ldots,j_n} u_{1,j_1} \cdots u_{n,j_n} \end{aligned}$$

Therefore, we have $f^\lambda([x_1, \ldots, x_n]) = \Sigma f([x_1, \ldots, x_n])$ for all $[x_1, \ldots, x_n] \in TX$.

We can apply $(-)^T$ to both sides of the equation $f^\lambda = \Sigma f$ to get $(f^\lambda)^T = (\Sigma f)^T$. We have $(\Sigma f)^T = \Sigma f$ because Σf is a morphism of $\mathbf{SG} + \mathbf{CSG}$-models, and therefore preserves the operators of $\mathbf{SG} + \mathbf{CSG}$. Therefore, we have $f^{(ST)} = (f^\lambda)^T = \Sigma f$.

This shows that the monad structure on ST given by the adjunction $F_{\mathbf{SG+CSG}} \dashv U$ is the same as the one given by the distributive law λ. □

2.3 Monoids

Now that we have the free semigroup, we can easily construct the free monoid by adding a new constant (arity 0 operator) to the signature, and adding the corresponding axioms.

Free commutative monoids

Let X be a set. We consider the theory of commutative monoids $\mathbf{CM}$, with a signature Σ with two operators: $(+)$ of arity 2, and 0 of arity 0, and with the three axioms

$$a, b \vdash a + b = b + a$$
$$a, b, c \vdash a + (b + c) = (a + b) + c$$
$$a \vdash a + 0 = a = 0 + a$$

We pose[2]

$$SX = \{f : X \to \mathbb{N} \mid f \text{ has finite support}\}$$

There is no non-empty support condition as in the free commutative semigroup, which means that the function f can be the constant 0 function. This corresponds to the new constant 0 in the signature.

We pose

$$\mathbf{0} = \lambda x.0 \in SX$$

.

We keep the same definition for the operator + and for $\eta_{\mathbf{CM}}$ as in the free commutative semigroup.

We pose $F_{\mathbf{CM}}X = (SX, (+), \mathbf{0})$ where + and $\eta_{\mathbf{CM}}$ are defined as above.
We want to show that $F_{\mathbf{CM}}X$ is a free **CM**-model over X.

- ▷ $F_{\mathbf{CM}}X$ is a **CM**-model: the axioms of **CM** are valid in $F_{\mathbf{CM}}X$ because of the commutativity and associativity of addition in $\mathbb{N}$, and because of the existence of the constant function $\mathbf{0}$ which acts as the identity for +.
- ▷ Let $A = (\underline{A}, +_A, 0_A)$ be a **CM**-model over X with environment $\varphi_A : X \to U\,A$. We define $\Sigma\varphi_A : SX \to \underline{A}$ for all $f \in SX$ by

$$\Sigma\varphi_A(f) = \sum_{x \in X} f\ x \cdot \varphi_A(x)$$

We have $\Sigma\varphi_A(\mathbf{0}) = 0_A$.
For the sake of doing the proof at least once, we'll prove that $\Sigma\varphi_A$ is well defined on $SX \smallsetminus \{\mathbf{0}\}$.

Proposition 80

For all $f \in SX$, the sum $\sum_{x \in X} f\ x \cdot \varphi_A(x)$ is well defined.

Proof
Let $f \in SX \smallsetminus \{\mathbf{0}\}$. We note $\operatorname{Supp} f = \{x \in X \mid f(x) \neq 0\}$.

Since f has finite support, $\operatorname{Supp} f$ is a finite set. Moreover, since $f \neq \mathbf{0}$, $\operatorname{Supp} f$ is non-empty. Therefore, we can write $\operatorname{Supp} f = \{x_1, x_2, ..., x_n\}$ for some $n \geq 1$ and we have a bijection σ between $\{1, 2, ..., n\}$ and $\operatorname{Supp} f$ given by $i \mapsto x_i$.

Therefore, we can rewrite the sum as follows:

$$\begin{aligned}\sum_{x \in X} f\ x \cdot \varphi_A(x) &= \sum_{x \in \operatorname{Supp} f} f\ x \cdot \varphi_A(x) \\ &= \sum_{i=1}^{n} f\ \sigma(i) \cdot \varphi_A(\sigma(i))\end{aligned}$$

The definition is valid because $+_A$ is associative.

Let σ' be another bijection between $\{1, 2, ..., n\}$ and $\operatorname{Supp} f$. Let's show that we get the same value for the sum if we use σ' instead of σ. This is true because σ and σ' are bijections, so there exists a permutation π of $\{1, 2, ..., n\}$ such that $\sigma' = \sigma \circ \pi$. Therefore, we have:

$$\begin{aligned}\sum_{i=1}^{n} f\ \sigma'(i) \cdot \varphi_A(\sigma'(i)) &= \sum_{i=1}^{n} f\ \sigma(\pi(i)) \cdot \varphi_A(\sigma(\pi(i))) \\ &= \sum_{i=1}^{n} f\ \sigma(i) \cdot \varphi_A(\sigma(i)) && \text{by commutativity of } +_A\end{aligned}$$

[2] we may want to find a different name, as it may be confused with the one for the commutative semigroup

Therefore, the definition of $\Sigma\varphi_A(f)$ does not depend on the choice of the bijection between $\{1, 2, ..., n\}$ and $\operatorname{Supp} f$, and is well defined.

Now if $f = \mathbf{0}$, we have $\operatorname{Supp} f = \varnothing$, and the sum is the empty sum, which is defined to be 0_A. □

We can now prove that $\Sigma\varphi_A$ is the unique morphism h of **CM**-models from $F_{\mathbf{CM}}X$ to A such that the following diagram commutes:

$$\begin{tikzcd} X \arrow[r, "\eta_{\mathbf{CM}}"] \arrow[dr, "\varphi_A"'] & SX \arrow[d, "h"] \\ & U\,A \end{tikzcd}$$

- Preservation of the operator $+$: let $f, g \in SX$. We have:

$$\begin{aligned} \Sigma\varphi_A(f+g) &= \Sigma\varphi_A(\lambda x. f\ x + g\ x) \\ &= \sum_{x\in X} (f\ x + g\ x)\cdot\varphi_A(x) && \text{by definition} \\ &= \sum_{x\in X} f\ x\cdot\varphi_A(x) + \sum_{x\in X} g\ x\cdot\varphi_A(x) && \text{since both } f \text{ and } g \text{ have finite support} \\ &= \Sigma\varphi_A(f) + \Sigma\varphi_A(g) \end{aligned}$$

- Commutation of the diagram: $\eta_{\mathbf{CM}} = \eta_{\mathbf{CSG}}$, so the proof is the same as for the free commutative semigroup.
- Uniqueness: let $h : F_{\mathbf{CM}}X \to A$ be a morphism of **CM**-models such that the diagram above commutes. We want to show that $h = \Sigma\varphi_A$. Let $f \in SX$. We'll show this lemma first:

Lemma 81

For all $f \in SX$, we have $f = \sum_{x\in X} f\ x\cdot\eta_{\mathbf{CM}}(x)$.

Proof

Let $f \in SX$. We have $\operatorname{Supp} f = \{x \in X \mid f(x) \neq 0\}$. Since f has finite support, $\operatorname{Supp} f$ is a finite set. Moreover, since f is not necessarily non-zero, $\operatorname{Supp} f$ can be empty.

If $\operatorname{Supp} f$ is empty, then $f = \mathbf{0}$ and we have:

$$\sum_{x\in X} f\ x\cdot\eta_{\mathbf{CM}}(x) = \sum_{x\in X} 0\cdot\eta_{\mathbf{CM}}(x) = 0 = f$$

If $\operatorname{Supp} f$ is non-empty, we can write $\operatorname{Supp} f = \{x_1, x_2, ..., x_n\}$ for some $n \geq 1$. We'll prove by induction on n that $f = \sum_{x\in X} f\ x\cdot\eta_{\mathbf{CM}}(x)$.

* If $n = 1$, then $\operatorname{Supp} f = \{x_1\}$ and we have $f = f\ x_1\cdot\eta_{\mathbf{CM}}(x_1) = \sum_{x\in X} f\ x\cdot\eta_{\mathbf{CM}}(x)$.
* If $n > 1$, we have $\operatorname{Supp} f = \{x_1, x_2, ..., x_n\}$.

  We can write $f = f\ x_1\cdot\eta_{\mathbf{CM}}(x_1) + f'$ where $f' = \lambda x. \begin{cases} 0 & \text{if } x = x_1 \\ f\ x & \text{otherwise} \end{cases}$

  We have $\operatorname{Supp} f' = \{x_2, ..., x_n\}$, so by the induction hypothesis, we have $f' = \sum_{x\in X} f'\ x\cdot\eta_{\mathbf{CM}}(x)$. Therefore, we have:

$$\begin{aligned} f &= f\ x_1\cdot\eta_{\mathbf{CM}}(x_1) + f' \\ &= f\ x_1\cdot\eta_{\mathbf{CM}}(x_1) + \sum_{x\in X} f'\ x\cdot\eta_{\mathbf{CM}}(x) \\ &= f\ x_1\cdot\eta_{\mathbf{CM}}(x_1) + \sum_{x\in X} f\ x\cdot\eta_{\mathbf{CM}}(x) - f\ x_1\cdot\eta_{\mathbf{CM}}(x_1) && \text{by definition of } f' \\ &= \sum_{x\in X} f\ x\cdot\eta_{\mathbf{CM}}(x) \end{aligned}$$

□

We then have:

$$\begin{aligned} h(f) &= h\left(\sum_{x\in X} f\ x \cdot \eta_{\mathbf{CM}}(x)\right) \\ &= \sum_{x\in X} f\ x \cdot h(\eta_{\mathbf{CM}}(x)) \\ &= \sum_{x\in X} f\ x \cdot \varphi_A(x) && \text{since the diagram commutes} \\ &= \Sigma\varphi_A(f) \end{aligned}$$

$\boxed{\text{Therefore we proved that } F_{\mathbf{CM}}X \text{ is a free } \mathbf{CM}\text{-model over } X.}$

3 Distributive combination

3.1 Definitions

Definition 82 (Coproduct of signatures)
Given two signatures Σ_1 and Σ_2, we define their coproduct $\Sigma_1 + \Sigma_2$ as the signature with operators the disjoint union of the operators, but retaining their arity:

$$\Sigma_1 + \Sigma_2 = \{\iota_i f : n \mid (f : n) \in \Sigma_i, i = 1, 2\}$$

Given a Σ_i-term-in-context $n \vdash_{\Sigma_i} t$, we denote by $n \vdash_{\Sigma_1+\Sigma_2} \iota_i[t]$ the $\Sigma_1 + \Sigma_2$-term-in-context obtained by applying the injection ι_i to all the operators of t.

Similarly, given a Σ_i-equation in context $eq := (n \vdash_{\Sigma_i} t = t')$, we denote by $\iota_i[eq]$ the $\Sigma_1 + \Sigma_2$-equation in context $n \vdash_{\Sigma_1+\Sigma_2} \iota_i[t] = \iota_i[t']$.

Definition 83 (Distributive combination of theories)
Let $\mathfrak{T} : \mathcal{O} \to \mathcal{M}$ be a translation from an operad $\mathcal{O}$ to a theory $\mathcal{M}$ and $\mathcal{A}$ be any theory.

We define the *distributive combination of* $\mathfrak{T}$ *over* $\mathcal{A}$ to be the theory $\mathfrak{T} \triangleleft \mathcal{A}$ given by

$$\begin{aligned} \Sigma_{\mathfrak{T}\triangleleft A} &:= \Sigma_M + \Sigma_A \\ \mathrm{Ax}_{\mathfrak{T}\triangleleft A} &:= \iota_1[\mathrm{Ax}_{\mathcal{M}}] \cup \iota_2[\mathrm{Ax}_{\mathcal{A}}] \\ &\cup \left\{ \begin{array}{l} \langle x_i \rangle_{\substack{i=1 \\ i\neq i_0}}^{n}\ ,\ \mathbf{y} \vdash \mathfrak{T}f(x_1, \ldots, x_{i_0-1}, s\mathbf{y}, x_{i_0+1}, \ldots, x_n) \\ \qquad\qquad = s\,\langle \mathfrak{T}f(x_1, \ldots, x_{i_0-1}, \mathbf{y}_j, x_{i_0+1}, \ldots, x_n) \rangle_{j=1}^{|\mathbf{y}|} \end{array} \;\middle|\; \begin{array}{l} (f : n) \in O_n \\ (s : |\mathbf{y}|) \in \Sigma_{\mathcal{A}} \end{array} \right\} \end{aligned}$$

3.2 Combination of semigroups and commutative semigroups

Let's first start with a case of study. We'll take $\mathcal{A}$ to be commutative semigroups and $\mathcal{M}$ to be semigroups. We denote by $(+)$ the operator on $\mathcal{A}$, and $(\cdot)$ the operator on $\mathcal{M}$.

We define $\mathcal{O}$ by $\mathcal{O}_0 = \varnothing$ and $\mathcal{O}_n = \{x_1, \ldots, x_n \vdash x_1 \cdots x_n\}$ for $n \geqslant 1$. We will simply write $\mathcal{O}_n = \{\mu_n\}$.

Proposition 84

$\mathcal{O}$ is the smallest operad containing[a] $(\cdot)$.

[a] we may say that $\mathcal{O}$ is the operad spanned by $(\cdot)$

Proof

▷ **$\mathcal{O}$ is an operad.**

associativity comes from the associativity of $(\cdot)$ and the neutrality of the identity is trivial.

▷ **minimality condition.**

Let $\mathcal{O}'$ be an operad containing $(\cdot)$. We want to show that $\mathcal{O}$ is a sub-operad of $\mathcal{O}'$. We have $\mu_2 \in \mathcal{O}'_2$ since $\mathcal{O}'$ contains $(\cdot)$. We can then construct μ_n for all $n \geqslant 1$ by composition of μ_2 with itself, and using the identity for the case $n = 1$. Therefore, we have $\mu_n \in \mathcal{O}'_n$ for all $n \geqslant 1$, which means that $\mathcal{O}$ is a sub-operad of $\mathcal{O}'$.

□

We then define a translation $\mathfrak{T} : \mathcal{O} \to \mathcal{M}$ by interpreting each operator as the n-ary multiplication, i.e $\mathfrak{T}\mu_n(x_1, \dots, x_n) = x_1 \cdots x_n$.

Proposition 85

$\mathfrak{T} : \mathcal{O} \to \mathcal{M}$ is a surjective translation.

Proof

▷ **$\mathfrak{T}$ is a translation.**

– the identity condition is trivialy true.

– the translation respects composition thanks to the associativity of $(\cdot)$.

▷ **$\mathfrak{T}$ is surjective.**

Let $t \in \mathcal{M}_n$. Since $\mathcal{M}$ is the algebraic theory of semigroups, t is represented by a nonempty word in the variables $x_1, \dots, x_n$. Thus we may write

$$t = x_{i_1} x_{i_2} \cdots x_{i_m}$$

for some $m \geqslant 1$ and indices $i_j \in \{1, \dots, n\}$.

Consider the unique m-ary operation $\mu_m \in \mathcal{O}_m$. By definition of the translation,

$$\mathfrak{T}(\mu_m) = x_1 x_2 \cdots x_m.$$

We define a renaming $\rho : m \to n$ by

$$\rho(j) = i_j.$$

Applying this renaming to $\mathfrak{T}(\mu_m)$ gives

$$\mathfrak{T}(\mu_m)[\rho] = (x_1 x_2 \cdots x_m)[\rho] = x_{\rho(1)} x_{\rho(2)} \cdots x_{\rho(m)} = x_{i_1} x_{i_2} \cdots x_{i_m} = t.$$

Therefore every term $t \in \mathcal{M}_n$ is of the form $t = \mathfrak{T}\mu_m[\rho]$ for some renaming $\rho : m \to n$.

Hence the translation $\mathfrak{T} : \mathcal{O} \to \mathcal{M}$ is surjective.

□

Let's give an explicit description of the theory $\mathfrak{T} \lhd \mathcal{A}$.

- **Signature.** The signature $\Sigma_{\mathfrak{T} \lhd \mathcal{A}}$ contains the operators of $\mathcal{M}$ and the operators of $\mathcal{A}$. Therefore, it contains the operator $(\cdot)$ of arity 2 and the operator $(+)$ of arity 2.

- **Axioms.**

$$
\begin{gathered}
\mathcal{M} \text{ axioms} \\
x, y, z \vdash x \cdot (y \cdot z) = (x \cdot y) \cdot z \\
\mathcal{A} \text{ axioms} \\
x, y \vdash x + y = y + x \\
x, y, z \vdash x + (y + z) = (x + y) + z \\
\text{distributivity axioms} \\
x, y, z \vdash x \cdot (y + z) = x \cdot y + x \cdot z \\
x, y, z \vdash (x + y) \cdot z = x \cdot z + y \cdot z
\end{gathered}
$$

We define the functor $\underline{\ \ } : \mathbf{Multi}(\mathcal{T},\mathcal{U}) \to \langle \mathcal{U}, \prod \rangle$ such that the following diagram commutes:

$$
\begin{tikzcd}
\mathbf{Multi}(\mathcal{T},\mathcal{U}) \arrow[r, "\underline{\ \ }"] \arrow[d, "-_o"'] & \langle \mathcal{U}, \prod \rangle \arrow[d, "-_o"] \\
\mathbf{Mod}(\mathcal{T},\mathcal{U}) \arrow[r, "U"'] & \mathcal{U}
\end{tikzcd}
$$

This is the anaology in **Multicat** of the forgetful functor in **Cat**.

Now we'll try to give an explicite description of each element of the following pullback square:

$$
\begin{tikzcd}
\mathbf{Mod}(\mathfrak{T} \lhd \mathcal{A}, \mathbf{Set}) \arrow[r, "\underline{\ \ }_{\mathcal{O}\lhd\mathcal{A}}"] \arrow[d, "\underline{\ \ }_{\mathcal{M}}"'] \arrow[dr, phantom, "\lrcorner", very near start] & \mathbf{Mod}(\mathcal{O}, \mathbf{Multi}(\mathcal{A}, \mathbf{Set})) \arrow[d, "\mathbf{Mod}(\mathcal{O}{,}\underline{\ \ })"] \\
\mathbf{Mod}(\mathcal{M}, \mathbf{Set}) \arrow[r, "\mathbf{Mod}(\mathfrak{T}{,}\mathbf{Set})"'] & \mathbf{Mod}(\mathcal{O}, \langle \mathbf{Set}, \prod \rangle)
\end{tikzcd}
$$

For vertices:

- $\mathbf{Mod}(\mathfrak{T} \lhd \mathcal{A}, \mathbf{Set})$ is the category of $\mathfrak{T} \lhd \mathcal{A}$-models in **Set** (i.e with carrier sets in **Set**), i.e the category of sem-rgs.
- $\mathbf{Mod}(\mathcal{M}, \mathbf{Set})$ is the category of $\mathcal{M}$-models in **Set**, i.e the category of semigroups.
- $\mathbf{Mod}(\mathcal{O}, \langle \mathbf{Set}, \prod \rangle)$ is the category of $\mathcal{O}$-models in $\langle \mathbf{Set}, \prod \rangle$. This category coincides with the category of semigroups in **Set** (cf. example 9 of Ohad's notes).
- $\mathbf{Mod}(\mathcal{O}, \mathbf{Multi}(\mathcal{A}, \mathbf{Set}))$ is the category of $\mathcal{O}$-models in $\mathbf{Multi}(\mathcal{A}, \mathbf{Set})$.

  $\mathbf{Multi}(\mathcal{A}, \mathbf{Set})$ is the multicategory where:

  - objects are $\mathcal{A}$-models in **Set**, i.e commutative semigroups.
  - multi-morphisms are multi-$\mathcal{A}$-homomorphisms, i.e ?

  Similarly to the previous case, $\mathbf{Mod}(\mathcal{O}, \mathbf{Multi}(\mathcal{A}, \mathbf{Set}))$ coincides with the category of sem-rgs in **Set**, except that we cannot guarantee that the distributivity axioms hold in $\mathbf{Mod}(\mathcal{O}, \mathbf{Multi}(\mathcal{A}, \mathbf{Set}))$.

For arrows:

- $\underline{\ \ }_{\mathcal{O}\lhd\mathcal{A}}$ is such that for $A = (UA, (+), (\cdot)) \in \mathbf{Mod}(\mathfrak{T} \lhd \mathcal{A}, \mathbf{Set})$, $\underline{A}_{\mathcal{O}\lhd\mathcal{A}}$ consists of:
  - $\underline{A} = (UA, (+))$ the $\mathcal{A}$-model given by the interpretation of the operators of $\mathcal{A}$ in A.
  - $\underline{A}_{\mathcal{O}\lhd\mathcal{A}}[\![\mu_n]\!](x_1, \ldots, x_n) = x_1 \cdot \cdots \cdot x_n \in \underline{A}$
- $\underline{\ \ }_{\mathcal{M}\lhd}$ is the functor sending a $\mathfrak{T} \lhd \mathcal{A}$-model M to the $\mathcal{M}$-model in **Set** given by the interpretation of the operators of $\mathcal{M}$ in M, i.e the semigroup structure of M.

  For $A = (\underline{A}, (+), (\cdot)) \in \mathbf{Mod}(\mathfrak{T} \lhd \mathcal{A}, \mathbf{Set})$, we have $\underline{A}_{\mathcal{M}\lhd} = (\underline{A}, (\cdot))$. It is easy to check that this is indeed a $\mathcal{M}$-model in **Set** because the axioms of $\mathcal{M}$ are included in the axioms of $\mathfrak{T} \lhd \mathcal{A}$.

- $\mathbf{Mod}(\mathfrak{T}, \mathbf{Set})$ is defined in Proposition 56. We have

$$\left(\mathbf{Mod}(\mathfrak{T}, \langle \mathcal{C}, \textstyle\prod \rangle)A\right)[\![f]\!] \coloneqq A[\![\mathfrak{T}f]\!] \qquad \mathbf{Mod}(\mathfrak{T}, \langle \mathcal{C}, \textstyle\prod \rangle)h \coloneqq h$$

- $\mathbf{Mod}(\mathcal{O}, \underline{\ \ })$ is defined in Proposition 58. It forgets about the model structure for $\mathcal{A}$, keeping only its carrier set, therefore we are left with a $\mathcal{O}$-model in **Set**.

Remark : $\mathbf{Mod}(\mathfrak{T}, \mathbf{Set})$ is therefore an isomorphism of categories.

Proposition 86

The square above is a pullback square.

Proof

▷ **commutativity.**

Let $A \in \mathbf{Mod}(\mathfrak{T} \triangleleft \mathcal{A}, \mathbf{Set})$.

We clearly have $\underline{\mathbf{Mod}(\mathfrak{T}, \mathbf{Set})(\underline{A}_{\mathcal{M}\triangleleft})} = \underline{\mathbf{Mod}(\mathcal{O}, \underline{\ \ })(\underline{A}_{\mathcal{O}\triangleleft\mathcal{A}})}$, since both sides are equal to $\underline{A}$.

Moreover, for $f \in \mathcal{O}_n$, we have:

$$\begin{aligned}
\mathbf{Mod}(\mathfrak{T}, \mathbf{Set})(\underline{A}_{\mathcal{M}\triangleleft})[\![f]\!] &= \underline{A}_{\mathcal{M}\triangleleft}[\![\mathfrak{T}f]\!] && \text{by definition of } \mathbf{Mod}(\mathfrak{T}, \mathbf{Set}) \\
&= \underline{A}_{\mathcal{M}\triangleleft}[\![\mu_n]\!] && \text{since } \mathcal{O} \text{ is spanned by } \mu_n \\
&= \underline{A}_{\mathcal{O}\triangleleft\mathcal{A}}[\![\mu_n]\!] && \text{by definition of } \underline{\ \ }_{\mathcal{O}\triangleleft\mathcal{A}} \\
&= \mathbf{Mod}(\mathcal{O}, \underline{\ \ })(\underline{A}_{\mathcal{O}\triangleleft\mathcal{A}})[\![f]\!] && \text{by definition of } \mathbf{Mod}(\mathcal{O}, \underline{\ \ })
\end{aligned}$$

▷ **pullback.**

Let $\mathcal{C}$ be a category, $F : \mathcal{C} \to \mathbf{Mod}(\mathcal{M}, \mathbf{Set})$ and $G : \mathcal{C} \to \mathbf{Mod}(\mathcal{O}, \langle \mathbf{Set}, \prod \rangle)$ be two functors such that $\mathbf{Mod}(\mathfrak{T}, \mathbf{Set}) \circ F = \mathbf{Mod}(\mathcal{O}, \underline{\ \ }) \circ G$. We want to show that there exists a unique functor $H : \mathcal{C} \to \mathbf{Mod}(\mathfrak{T} \triangleleft \mathcal{A}, \mathbf{Set})$ such that the following diagram commutes:

$$\begin{tikzcd}
\mathcal{C} \arrow[dr, dashed, "H"] \arrow[drr, bend left, "G"] \arrow[ddr, bend right, "F"'] & & \\
& \mathbf{Mod}(\mathfrak{T} \triangleleft \mathcal{A}, \mathbf{Set}) \arrow[r, "\underline{\ \ }_{\mathcal{O}\triangleleft\mathcal{A}}"] \arrow[d, "\underline{\ \ }_{\mathcal{M}}"'] & \mathbf{Mod}(\mathcal{O}, \mathbf{Multi}(\mathcal{A}, \mathbf{Set})) \arrow[d, "\mathbf{Mod}(\mathcal{O}, \underline{\ \ })"] \\
& \mathbf{Mod}(\mathcal{M}, \mathbf{Set}) \arrow[r, "\mathbf{Mod}(\mathfrak{T}, \mathbf{Set})"'] & \mathbf{Mod}(\mathcal{O}, \langle \mathbf{Set}, \prod \rangle)
\end{tikzcd}$$

We define $H : \mathcal{C} \to \mathbf{Mod}(\mathfrak{T} \triangleleft \mathcal{A}, \mathbf{Set})$ as follows: for $c \in \mathcal{C}$, we have $Hc = (UFc, (+_c), (\cdot_c))$ where:

- UFc is the underlying set of the $\mathcal{M}$-model Fc.
- $(+_c)$ is the binary operation on UFc given by the interpretation of the operator $(+)$ of $\mathcal{A}$ in Gc, i.e $(+_c) = Gc[\![+]\!]$.
- $(\cdot_c)$ is the binary operation on UFc given by the interpretation of the operator $(\cdot)$ of $\mathcal{M}$ in Fc, i.e $(\cdot_c) = Fc[\![\cdot]\!]$.

We then have to check that Hc is indeed a $\mathfrak{T} \triangleleft \mathcal{A}$-model, i.e that the axioms of $\mathfrak{T} \triangleleft \mathcal{A}$ hold in Hc. This is true because:

- the axioms of $\mathcal{M}$ hold in Hc since Fc is a $\mathcal{M}$-model and the interpretation of the operators of $\mathcal{M}$ in Hc is the same as the interpretation of the operators of $\mathcal{M}$ in Fc.

- the axioms of $\mathcal{A}$ hold in Hc since Gc is a $\mathcal{O}$-model in $\mathbf{Multi}(\mathcal{A}, \mathbf{Set})$ and the interpretation of the operators of $\mathcal{A}$ in Hc is the same as the interpretation of the operators of $\mathcal{A}$ in Gc.
- the distributivity axioms ...

$\square$

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A Daily recap

- 05/05/2026: first day, talk with Ohad, building tour, admin stuff, setting up the git repo and starting to read the notes of Ohad

- 06/05/2026: reading Ohad's notes, starting to write my own notes. lots of definitions to try and understand how things work.

- 07/05/2026: read the paper on frexes and frals, wrote down the relevant definitions. Meeting with Ohad: 3 equivalent definitions of adjunctions ⇒ universal arrows, examples where/when they are used + equivalence of definitions of free algebras and how to construct a free algebra for (commutative) semigroups.

  Gave examples of uses of universal arrows: defining adjunctions, defining products and coproducts, and defining free algebras.

  Defined and proved the free algebra for commutative semigroups, started to work on the free algebra for semigroups.

- 08/05/2026: Defined and proved the free algebra for semigroups and commutative monoids. Proved the details that were missing for the free algebra for commutative semirings (in the commutative monoid part).

  Started to work on the free algebra for the combination of semigroups and commutative semigroups, but still need to define the product operator and check that the distributivity axioms hold, and that the resulting algebra is indeed a free algebra for the combination.

- 11/05/2026: finished the definition of the product operator for the free algebra for the combination of semigroups and commutative semigroups, and proved that it is indeed a free algebra for the combination.

  Defined multiplication for the free semigroup and commutative semigroup, and proved that we get a monad structure on **Set** in both cases.

  Defined the distributive law of the monad for the free semigroup over the monad for the free commutative semigroup, and proved that it is indeed a distributive law → we get a monad structure on **Set** for the free algebra for the combination of semigroups and commutative semigroups.

- 12/05/2026: definition of Kleisli category and related objects, research and definition of algebra and distributive laws of monads in extension form, and link with the previous definition of distributive laws. Example 8.5 of [MW10] is almost the same thing as Section 2.2.

- 13/05/2026: started looking into the distributive combination of semigroups and commutative semigroups. Defined the operad $\mathcal{O}$ spanned by the multiplication operator of semigroups, and defined a translation from $\mathcal{O}$ to the theory of semigroups. Proved that $\mathcal{O}$ is minimal and that the translation is surjective.

- 14/05/2026: defined the monad on $\mathbf{SG} + \mathbf{CSG}$ with the adjunction, and proved that it is the same as the one given by the distributive law.

  Copied over relevant information to start describing the pullback square for the combination of models in more details. A lot of reading Ohad's notes.

- 15/05/2026: more copying definitions to understand the pullback square. Figured out what all of the objects (both vertices and arrows) are. 2-hour meeting with Ohad to check that I got everything right, and to talk about what I have to do next.

- 18/05/2026: finished giving the full description of all objects in the pullback square.

  Jesse helped me understand how to create the pullback in the category of categories, and started working on the proof that the square is a pullback square. Proved the commutativity of the square, and started working on the pullback condition. It seems like the distributivity axioms pose a problem.

B TODO

- follow Ohad's abstract picture
- dig in the paper about frexes to learn more about that (both frexes and frals) and write the relevant definitions down
- write the definition of free extensions with universal arrows
- look at what was done with involution to take inspiration

C Questions for Ohad

- what is the reader monad